Sharp first-order asymptotics for the squared Lagrange interpolation integral

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Version 1, 24 October 2026

Abstract

For n distinct nodes in $[-1,1]$, consider the ordinary integral of the sum of the squares of their fundamental Lagrange polynomials, with all cardinal weights equal to one. We determine the sharp first-order asymptotic of its infimum:

$$m_n = 2 - \frac{c_*}{n} + o(n^{-1}), \qquad c_* = \frac{\Gamma_{\mathrm{E}}(1/4)^2}{8\Gamma_{\mathrm{E}}(3/4)^2} = 1.0942198076\ldots.$$

The lower bound holds for arbitrary node configurations, including endpoint nodes, without a separation or quadrature hypothesis. Zeros of the Jacobi polynomial $P_n^{(-3/4,-3/4)}$ give the matching upper bound. The proof retains the interpolation residual in the Legendre reproducing-kernel inequality. Its limiting contribution is a Hardy-space penalty, which combines with the scalar and phase corrections through a positive two-point entropy identity. The result determines the leading coefficient and the full limiting asymptotic; it does not determine the minimum for each fixed n.

1 Introduction

Let $X = \{x_1, \ldots, x_n\} \subset [-1,1]$ consist of n distinct real nodes, and let

$$\ell_i^X(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}, \qquad I_n(X) = \int_{-1}^{1} \sum_{i=1}^{n} \ell_i^X(x)^2 \, dx, \qquad m_n = \inf_{\substack{X \subset [-1,1] \\ |X| = n}} I_n(X). \tag{1.1}$$

The measure in this definition is ordinary, unnormalized Lebesgue measure. There is no choice of cardinal weights in the infimum. For $n = 1$ the single fundamental polynomial is 1, so $m_1 = 2$.

Erdős asked for the minimum of this integral and, in particular, whether $m_n = 2 - (1 + o(1))/n$; see the problem record [5] and the early study [1]. Brutman and Toledano [2] subsequently studied the same functional by numerical optimization and Richardson extrapolation. They predicted the leading coefficient $1.094219\ldots$ in the expansion of m_n [2, pp. 43–45, Eq. (28), Tables 1–2]. Their asymptotic conclusion assumes an expansion and is supported by finite computations, rather than by a sharp lower bound for arbitrary node sets. They also introduced an explicit "Average" set, obtained similar leading digits, and conjectured its asymptotic optimality [2, pp. 46–47, Eqs. (33), (35)–(36)].

The coefficient-one conjecture also has an explicit public disproof: the construction of `seanm27lol` [3] gives node sets whose scaled deficit is eventually at least $106/105$. That argument does not determine the sharp coefficient or convergence of $n(2 - m_n)$. The recent preprint [4] develops a different asymptotic framework with conditional endpoint assumptions.

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We prove the sharp first-order limit with an exact gamma-quotient constant, in agreement with the numerical prediction of Brutman and Toledano. The lower bound holds for arbitrary distinct nodes. Its matching Jacobi construction differs from their Average set; their separate conjecture for that set is not asserted here. No first-priority claim for the constant or the construction is made.

For $s > 0$, write $\Gamma_{\mathrm{E}}(s) = \int_0^\infty t^{s-1}e^{-t}\,dt$ for the Euler gamma function. The main result concerns the full asymptotic along all positive integers.

Theorem 1.1. *Define m_n by (1.1) and put*

$$c_* = \frac{\Gamma_{\mathrm{E}}(1/4)^2}{8\Gamma_{\mathrm{E}}(3/4)^2}. \tag{1.2}$$

Then

$$\lim_{n\to\infty} n(2 - m_n) = c_*. \tag{1.3}$$

More precisely, for every sequence of node sets $X_n \subset [-1,1]$ with $|X_n| = n$ and distinct nodes,

$$\limsup_{n\to\infty} n(2 - I_n(X_n)) \le c_*. \tag{1.4}$$

If X_n^{J} is the set of zeros of the Jacobi polynomial $P_n^{(-3/4,-3/4)}$, then

$$\lim_{n\to\infty} n(2 - I_n(X_n^{\mathrm{J}})) = c_*. \tag{1.5}$$

All nodes in this matching construction lie in $(-1,1)$.

The universal assertion includes configurations with endpoint nodes and with arbitrarily small gaps. It does not assume symmetry, stationarity, positive quadrature weights, or attainment of the infimum. The theorem does not identify the exact minimum or minimizing configurations at a fixed degree.

The difficulty in the lower bound is the interpolation constraint. A reproducing-kernel estimate gives a positive residual in addition to its Christoffel term. Dropping that residual loses the sharp first-order information. We retain it and use finitely many of the highest Legendre modes to obtain a limiting nonnegative penalty. At the same time, the node polynomial defines a zero-free analytic factor on the disk. The actual interpolation Gram matrix identifies the boundary phase of this factor. A two-point entropy identity then compares the scalar correction, the phase correction, and the residual penalty in one expression. The remaining loss is nonnegative, and its zero case gives the value in (1.2).

Section 2 establishes estimates uniform over the original node configurations. Section 3 extracts the analytic factor and identifies its phase. Section 4 expresses the first-order deficit in terms of this factor. Section 5 retains the residual and proves the sharp entropy comparison. Section 6 evaluates the actual Jacobi construction. The final section combines these statements on common subsequences and then passes to near-minimizers.

Conventions

The ordinary Legendre polynomial L_j is normalized by $L_j(1) = 1$. Its $L^2([-1,1], dx)$ normalization and the degree-$(n-1)$ reproducing kernel are

$$\psi_j(x) = \sqrt{j + \tfrac{1}{2}}\,L_j(x), \qquad K_n(x,t) = \sum_{j=0}^{n-1} \psi_j(x)\psi_j(t), \qquad \lambda_n(t) = \frac{1}{K_n(t,t)}. \tag{1.6}$$

Inner products of real polynomials use ordinary dx. Circle Hardy norms use normalized angular measure $d\theta/(2\pi)$ on the full circle; the upper-half angular probability measure is $d\theta/\pi$. The

disk is $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. All fractional analytic powers used below are normalized to equal 1 at $z = 0$. A boundary phase is the imaginary part of the normalized analytic logarithm, rather than a choice of principal argument. The nonnegative residual penalty will be denoted by $\mathcal{G}$, to distinguish it from Γ_{E}.

2 Uniform estimates for sublevel configurations

2.1 A uniform finite bound

Throughout this section, $n \geq 2$, the nodes $x_1, \ldots, x_n$ are distinct and belong to $[-1, 1]$, and every interval norm and integral uses ordinary, unnormalized Lebesgue measure dx. Write

$$I = \int_{-1}^{1} \sum_{i=1}^{n} \ell_i(x)^2 \, dx, \qquad s_i = \sqrt{1 - x_i^2}, \qquad D = 49 + 8\pi.$$

We shall prove

$$I \geq 2 - \frac{D}{2n} + \frac{1}{n(2n-1)}. \tag{2.1}$$

The two estimates needed for this purpose are

$$I \leq 2 \quad \Longrightarrow \quad \sum_i s_i \geq \frac{2n - 49}{\pi} \tag{2.2}$$

and the uniform Christoffel bound

$$\lambda_n(x) \geq \frac{\pi\sqrt{1-x^2}}{n} - \frac{8\pi}{n^2}, \qquad -1 \leq x \leq 1. \tag{2.3}$$

Both estimates include endpoint nodes.

2.1.1 Polynomial identities

We first record the normalizations and finite identities used below. Define

$$L_k(x) = \frac{1}{2^k k!} \frac{d^k}{dx^k}(x^2 - 1)^k.$$

This polynomial has parity k, leading coefficient $\gamma_k = 2^{-k}\binom{2k}{k}$, and $L_k(1) = 1$. For every polynomial p of degree less than k, integration by parts k times gives $\int_{-1}^{1} L_k p \, dx = 0$; all boundary terms vanish because $(x^2 - 1)^k$ has zeros of multiplicity k at both endpoints. If $B_k = \int_{-1}^{1} (1 - x^2)^k \, dx$, integration of $[x(1 - x^2)^k]'$ gives

$$B_0 = 2, \qquad (2k+1)B_k = 2kB_{k-1}, \qquad B_k = \frac{2^{2k+1}(k!)^2}{(2k+1)!}.$$

The same integrations by parts yield $\int L_k x^k \, dx = 2^{-k} B_k$. Orthogonality to the lower-degree terms therefore proves

$$\int_{-1}^{1} L_k(x)^2 \, dx = \frac{2}{2k+1}, \qquad \psi_k = \sqrt{k + \tfrac{1}{2}}\, L_k. \tag{2.4}$$

Expanding xL_k in this orthogonal basis eliminates degrees at most $k - 2$ by orthogonality and degree k by parity. Its leading coefficient and its value at 1 determine the remaining two coefficients. Thus

$$(2k+1)xL_k = (k+1)L_{k+1} + kL_{k-1}, \qquad L_0 = 1, \quad L_1 = x, \tag{2.5}$$

where the $k=0$ formula omits L_{-1}. The operator $\mathcal{L}p = -[(1-x^2)p']'$ preserves degree and has leading multiplier $k(k+1)$ on degree k. Integration by parts shows that $\mathcal{L}L_k$ is orthogonal to every lower-degree polynomial. Comparing leading coefficients proves

$$[(1-x^2)L_k']' + k(k+1)L_k = 0. \tag{2.6}$$

Similarly, $Q = (1-x^2)L_k' + kxL_k$ has degree at most $k-1$ and, for $\deg p \le k-2$,

$$\int_{-1}^{1} Qp\,dx = \int_{-1}^{1} L_k\{(k+2)xp - (1-x^2)p'\}\,dx = 0.$$

Its value at 1 consequently gives the lowering identities

$$(1-x^2)L_n' = n(L_{n-1} - xL_n), \qquad (1-x^2)L_{n-1}' = n(xL_{n-1} - L_n). \tag{2.7}$$

The second formula follows from the first at degree $n-1$ and (2.5).

Subtracting the recurrence at two arguments and summing the resulting telescoping terms gives

$$K_n(x,y) = \frac{n}{2}\,\frac{L_n(x)L_{n-1}(y) - L_{n-1}(x)L_n(y)}{x-y}. \tag{2.8}$$

The diagonal is removable and equals

$$K_n(x,x) = \frac{n}{2}\{L_n'(x)L_{n-1}(x) - L_n(x)L_{n-1}'(x)\}.$$

The lowering identities now give, with $h^2 = 1-x^2 > 0$,

$$\begin{aligned} 2K_n(x,x) &= \frac{n^2}{h^2}\{L_{n-1}^2 - 2xL_nL_{n-1} + L_n^2\} \\ &= h^2L_n'(x)^2 + n^2L_n(x)^2 \\ &= h^2L_{n-1}'(x)^2 + n^2L_{n-1}(x)^2. \end{aligned} \tag{2.9}$$

The last two expressions are polynomial identities, so they extend to the endpoints. Directly from the defining sum, $K_n(\pm1,\pm1) = n^2/2$.

We also need an elementary central-binomial estimate. Put $C_m = 4^{-m}\binom{2m}{m}$ and $J_r = \int_0^{\pi/2}\sin^r t\,dt$. Integration by parts gives

$$J_r = \frac{r-1}{r}J_{r-2}, \qquad J_{2m} = \frac{\pi}{2}C_m, \qquad J_{2m-1} = \frac{1}{2mC_m}.$$

Since $J_{2m} \le J_{2m-1}$,

$$C_m^2 \le \frac{1}{\pi m}, \qquad m \ge 1. \tag{2.10}$$

The recurrence and lowering formula also imply

$$L_{2m}(0) = (-1)^mC_m, \qquad L_{2m+1}(0) = 0, \qquad L_{2m+1}'(0) = (2m+1)(-1)^mC_m. \tag{2.11}$$

2.1.2 An averaged endpoint logarithm

For this and the following two subsections suppose $I \le 2$. Set

$$Q(x) = \prod_i (x-x_i), \quad b_i = \frac{1}{Q'(x_i)}, \quad H_0 = \sum_i b_i^2, \quad L_n^{\mathrm{cut}}(\theta) = \log\frac{2}{\sin\theta + 1/n}.$$

We claim

$$H_0 \le 4^n, \qquad \sum_i L_n^{\mathrm{cut}}(\arccos x_i) \le 3n. \tag{2.12}$$

For the first assertion use the second-kind Chebyshev polynomials $U_k(\cos\theta) = \sin((k+1)\theta)/\sin\theta$. Their recurrence $U_{k+1} = 2xU_k - U_{k-1}$ gives leading coefficient 2^k, and sine orthogonality gives

$$\int_{-1}^{1} U_j(x)U_k(x)\sqrt{1-x^2}\,dx = \frac{\pi}{2}\delta_{jk}.$$

Consequently, if p has degree at most $N = n-1$ and leading coefficient a at degree N, then

$$a = \frac{2^{N+1}}{\pi}\int_{-1}^{1} p(x)U_N(x)\sqrt{1-x^2}\,dx, \qquad a^2 \le \frac{2^{2N+3}}{\pi^2}\int_{-1}^{1} p(x)^2\,dx.$$

Here Cauchy–Schwarz uses $U_N(x)^2(1-x^2) = \sin^2(n\arccos x) \le 1$. Apply this to each cardinal polynomial, whose leading coefficient is b_i. Summing gives

$$H_0 \le \frac{2^{2n+1}}{\pi^2} I \le \frac{4}{\pi^2}4^n \le 4^n.$$

Let $z = \sqrt{1+n^{-2}} > 1$. The identity $\sum_i \ell_i = 1$ gives

$$1 = Q(\pm z)\sum_i \frac{b_i}{\pm z - x_i}.$$

Since $z - 1 = 1/[n^2(z+1)] \ge 1/(3n^2)$, Cauchy–Schwarz implies

$$|Q(z)|,\ |Q(-z)| \ge \frac{z-1}{\sqrt{nH_0}} \ge \frac{1}{3n^{5/2}2^n}.$$

It follows that

$$\prod_i (s_i^2 + n^{-2}) = |Q(z)Q(-z)| \ge \frac{1}{9n^5 4^n}.$$

Because $s_i + n^{-1} \ge \sqrt{s_i^2 + n^{-2}}$, taking logarithms yields

$$\sum_i \log\frac{2}{s_i + 1/n} \le 2n\log 2 + \frac{5}{2}\log n + \log 3.$$

For $n \ge 2$, division by n and the inequalities $\log n/n \le 1/e$, $\log 2 < 7/10$, $\log 3 < 6/5$, and $e > 8/3$ bound the right side by $7/5 + 15/16 + 3/5 = 47/16 < 3$. This proves (2.12). In particular its cutoff makes the estimate valid at $s_i = 0$.

2.1.3 The information matrix and its kernel

Use the basis orthonormal for the arcsine probability measure $dx/(\pi\sqrt{1-x^2})$:

$$\phi_0 = 1, \qquad \phi_k = \sqrt{2}\,T_k \quad (1 \le k < n), \qquad \phi = (\phi_0, \dots, \phi_{n-1})^T.$$

Define

$$V_{ik} = \phi_k(x_i), \qquad M = \frac{1}{n}V^T V, \qquad A = \int_{-1}^{1}\phi(x)\phi(x)^T\,dx, \qquad F_n(x) = \frac{1}{n}\phi(x)^T A\phi(x).$$

Distinct nodes make V invertible. Exact interpolation and finite trace identities give

$$nI = \operatorname{tr}(AM^{-1}), \qquad \sum_i F_n(x_i) = \operatorname{tr}(AM). \tag{2.13}$$

For positive M, the scalar inequality $\mu + \mu^{-1} \geq 2$ on its eigenvalues gives $M + M^{-1} \succeq 2\mathrm{Id}$. Taking its trace against the positive matrix A is valid without assuming commutativity. Hence

$$\sum_i F_n(x_i) \geq 2 \operatorname{tr} A - nI. \tag{2.14}$$

A finite cosine integral gives

$$\int_{-1}^{1} 2T_k(x)^2 \, dx = 2 - \frac{2}{4k^2 - 1}, \qquad k \geq 1.$$

The corrections telescope, since $2/(4k^2 - 1) = 1/(2k - 1) - 1/(2k + 1)$. Therefore

$$\operatorname{tr} A = 2n - 1 + \frac{1}{2n - 1}, \qquad \sum_i F_n(x_i) \geq 2n - 2 + \frac{2}{2n - 1} \quad (I \leq 2). \tag{2.15}$$

We next prove, uniformly for $0 \leq \theta \leq \pi$,

$$F_n(\cos\theta) \leq \pi \sin\theta + \frac{\pi^2 L_n^{\mathrm{cut}}(\theta) + C_0}{n} \leq \pi \sin\theta + \frac{10 L_n^{\mathrm{cut}}(\theta) + 17}{n}, \tag{2.16}$$

where $C_0 = (\pi^2/2)(1 + \log(2\pi)) + \pi/2 < 17$. Let $m = 2n - 1$ and

$$\mathcal{D}(u) = \sum_{k=-(n-1)}^{n-1} e^{iku} = \frac{\sin(mu/2)}{\sin(u/2)}, \qquad w(u) = |\sin u|,$$

with removable values filled in. The arcsine reproducing kernel is

$$\mathcal{K}(\theta, \varphi) = 1 + 2 \sum_{k=1}^{n-1} \cos(k\theta) \cos(k\varphi) = \frac{\mathcal{D}(\theta - \varphi) + \mathcal{D}(\theta + \varphi)}{2}.$$

Changing variables $x = \cos\varphi$ and extending evenly to the circle gives the exact decomposition $F_n(\cos\theta) = T(\theta) + C(\theta)$, where

$$T(\theta) = \frac{1}{4n} \int_{-\pi}^{\pi} \mathcal{D}(\theta - \varphi)^2 w(\varphi) \, d\varphi,$$
$$C(\theta) = \frac{1}{4n} \int_{-\pi}^{\pi} \mathcal{D}(\theta - \varphi) \mathcal{D}(\theta + \varphi) w(\varphi) \, d\varphi.$$

These are finite trigonometric polynomials.

For the square term put

$$g(\theta) = \frac{1}{2\pi m} \int_{-\pi}^{\pi} \mathcal{D}(\delta)^2 w(\theta - \delta) \, d\delta, \qquad d = \min(\theta, \pi - \theta).$$

Since $\int_{-\pi}^{\pi} \mathcal{D}^2 = 2\pi m$, we have $T = \pi m g/(2n)$. Symmetrizing in δ shows

$$g(\theta) - \sin\theta \leq \frac{1}{\pi m} \int_{d}^{\pi} \mathcal{D}(\delta)^2 \delta \, d\delta. \tag{2.17}$$

Indeed reflection reduces the check to $0 \leq \theta \leq \pi/2$. For $0 \leq \delta \leq \theta$ the symmetrized difference is $\sin\theta(\cos\delta - 1) \leq 0$; for $\theta \leq \delta \leq \pi - \theta$ it is $\cos\theta \sin\delta - \sin\theta \leq \delta$; and for $\pi - \theta \leq \delta \leq \pi$ it is $-\sin\theta(\cos\delta + 1) \leq 0$. For $0 < \delta \leq \pi$, $|\mathcal{D}(\delta)| \leq \min(m, \pi/\delta)$. Splitting at π/m gives

$$\int_{d}^{\pi} \mathcal{D}(\delta)^2 \delta \, d\delta \leq \pi^2 \left\{ \frac{1}{2} + \log \frac{2\pi}{d + \pi/m} \right\}.$$

For $d \ge \pi/m$ the bound before this simplification is $\pi^2 \log(\pi/d)$, and for $d \le \pi/m$ it is $\pi^2(1/2+\log m)$; use $\max(d, \pi/m) \ge (d+\pi/m)/2$ in both cases. Since $d \ge \sin\theta$ and $\pi/m \ge 1/n$, (2.17) implies

$$T(\theta) \le \pi \sin\theta + \frac{\pi^2}{2n}\{L_n^{\mathrm{cut}}(\theta) + \tfrac{1}{2} + \log\pi\}. \tag{2.18}$$

For the cross term, the finite identity

$$\int_{-\pi}^{\pi} \mathcal{D}(\theta-\varphi)\mathcal{D}(\theta+\varphi)\, d\varphi = 2\pi\mathcal{D}(2\theta)$$

gives

$$C(\theta) = \frac{\pi}{2n}\sin\theta\, \mathcal{D}(2\theta) + R(\theta).$$

The first term has absolute value at most $\pi/(2n)$, because $\sin\theta\, \mathcal{D}(2\theta) = \sin(m\theta)$. In R, the weight is $w(\varphi) - \sin\theta$. For $0 < \theta < \pi$, the sine and cosine difference formulas give, for $0 \le \varphi \le \pi$,

$$|\mathcal{D}(\theta-\varphi)\mathcal{D}(\theta+\varphi)||\sin\varphi - \sin\theta| \le 2\left|\cot\frac{\theta+\varphi}{2}\right|.$$

Here the denominator product is $|\cos\varphi - \cos\theta|/2$, and the canceled quotient equals $|\cot((\theta+\varphi)/2)|$. Thus

$$|R(\theta)| \le \frac{1}{n}\int_0^{\pi}\left|\cot\frac{\theta+\varphi}{2}\right| d\varphi = \frac{2}{n}\log\frac{2}{\sin\theta}.$$

There is also a bound finite at the endpoints. The inequality $2|ab| \le a^2 + b^2$, evenness, and the circle Lipschitz bound for w give

$$|R(\theta)| \le \frac{1}{4n}\int_{-\pi}^{\pi} \mathcal{D}(\delta)^2|\delta|\, d\delta \le \frac{\pi^2}{2n}(\tfrac{1}{2} + \log m).$$

If $\sin\theta \ge 1/n$, use the preceding logarithmic bound and $\log(2/\sin\theta) \le L_n^{\mathrm{cut}}(\theta) + \log 2$. If $\sin\theta \le 1/n$, use $L_n^{\mathrm{cut}}(\theta) \ge \log n$ and $\log m \le \log n + \log 2$. At the endpoints use the finite bound itself. Since $\pi^2/2 > 2$ and $L_n^{\mathrm{cut}} \ge 0$, all cases yield

$$|R(\theta)| \le \frac{\pi^2}{2n}\{L_n^{\mathrm{cut}}(\theta) + \tfrac{1}{2} + \log 2\}.$$

Combining this with (2.18) proves (2.16); its last inequality follows from $\pi^2 < 10$, $\log(2\pi) < 2$, and $\pi/2 < 2$.

Summing (2.16) at the nodes and using (2.12) now gives

$$\sum_i F_n(x_i) \le \pi \sum_i s_i + 47. \tag{2.19}$$

Together with (2.15), this proves $\pi\sum_i s_i \ge 2n - 49 + 2/(2n-1)$ and hence (2.2).

2.1.4 A Christoffel estimate valid up to the endpoints

For $x = \cos\theta$, $h = \sin\theta > 0$, put

$$u(\theta) = \sqrt{h}\, L_n(\cos\theta), \qquad q(\theta) = (n+\tfrac{1}{2})^2 + \frac{1}{4h^2}.$$

Equation (2.6) gives $u'' + qu = 0$. Therefore

$$E = u^2 + \frac{u'^2}{q}, \qquad E' = -\frac{q'u'^2}{q^2}.$$

The function q decreases on $(0,\pi/2)$ and increases on $(\pi/2,\pi)$, so $E(\theta)\le E(\pi/2)=c_n$ throughout the open interval. By (2.11),

$$c_{2m}=C_m^2,\qquad c_{2m+1}=\frac{(2m+1)^2C_m^2}{(2m+1)^2+(2m+1)+1/2}.$$

The bound (2.10) implies, for every $n\ge 2$,

$$c_n\le\frac{2}{\pi n}+\frac{4}{\pi n^2}. \tag{2.20}$$

For even n the second term is unnecessary; for odd $n\ge 3$, use $c_n\le 2/[\pi(n-1)]$ and $1/[n(n-1)]\le 2/n^2$.

In terms of u, (2.9) reads

$$K_n(\cos\theta,\cos\theta)=\frac{(u'-(\cot\theta/2)u)^2+n^2u^2}{2h}.$$

Let $a=|\cot\theta|/2$. Relative to $(u,u'/\sqrt{q})$, the numerator has larger diagonal entry q and off-diagonal entry of absolute value $a\sqrt{q}$, because $q=n^2+n+1/2+a^2$. It is consequently at most

$$(q+a\sqrt{q})E\le\{n^2+n+\tfrac12+2a^2+a(n+1)\}E,$$

where $\sqrt{q}\le n+1+a$ was used. On $nh\ge 1$, the inequalities $a\le 1/(2h)$, $h\le 1$, and $n\ge 2$ give

$$n+\tfrac12+2a^2+a(n+1)\le\frac{3n}{h}.$$

Using (2.20) thus proves

$$K_n(\cos\theta,\cos\theta)\le\frac{n}{\pi h}+\frac{2}{\pi h}+\frac{3}{\pi h^2}+\frac{6}{\pi nh^2}\le\frac{n}{\pi h}+\frac{8}{\pi h^2}.$$

Inverting this positive upper bound yields

$$\lambda_n(\cos\theta)\ge\frac{\pi h^2}{nh+8}=\frac{\pi h}{n}-\frac{8\pi h}{n(nh+8)}\ge\frac{\pi h}{n}-\frac{8\pi}{n^2}.$$

On $nh<1$ the right side of (2.3) is negative, so positivity suffices. The same argument applies at $h=0$ using the finite positive endpoint kernel. This proves (2.3) on the whole closed interval.

2.1.5 Combining the two bounds

Reproduction and Cauchy–Schwarz show

$$1=\ell_i(x_i)^2\le K_n(x_i,x_i)\|\ell_i\|^2_{L^2(dx)}.$$

Hence

$$I\ge\sum_i\lambda_n(x_i)\ge\frac{\pi}{n}\sum_i s_i-\frac{8\pi}{n}. \tag{2.21}$$

Suppose first $I\le 2$. Keeping the common trace in (2.14) and (2.19), and then using (2.21), gives

$$2\operatorname{tr}A-nI\le\operatorname{tr}(AM)\le\pi\sum_i s_i+47\le nI+8\pi+47.$$

Substitution of (2.15) proves (2.1). If $I>2$, its right side is less than 2 for every $n\ge 2$, and the conclusion follows immediately. No node separation beyond distinctness has entered the proof.

2.2 Uniform interpolation and angular moments

Normalize the nodal polynomial by

$$q_n(x) = 2^{n-1} \prod_{i=1}^{n} (x - x_i).$$

Retain $D = 49 + 8\pi$ and define the absolute constants

$$H = \frac{\sqrt{D} + \sqrt{D+8}}{2}, \qquad B = \sqrt{2} + H, \qquad C = \log\left(\frac{4 + 8B^2}{\pi}\right). \tag{2.22}$$

For every $n \geq 2$ and every distinct node set with $I_n \leq 2$, we shall prove

$$\left\| \sum_i a_i \ell_i \right\|_{L^2(dx)} \leq H \max_i |a_i| \qquad (a_i \in \mathbb{R}), \tag{2.23}$$

and

$$\|q_n\|_{L^2(dx)} \leq B, \qquad \left| \sum_i T_k(x_i) \right| \leq Ck \qquad (k \geq 1). \tag{2.24}$$

The latter estimate holds simultaneously for all positive integer k.

2.2.1 The Christoffel excess

Write $\lambda_i = \lambda_n(x_i)$, $L_X = \sum_i \lambda_i$, and $d_n = I_n - L_X$. The bounds (2.2) and (2.3), together with (2.21), imply

$$2 - \frac{D}{n} \leq L_X \leq I_n \leq 2, \qquad 0 \leq n d_n \leq D. \tag{2.25}$$

Let $K_i = K_n(\cdot, x_i)$ and $r_i = \ell_i - \lambda_i K_i$. Reproduction gives

$$\|r_i\|^2_{L^2(dx)} = \|\ell_i\|^2_{L^2(dx)} - \lambda_i, \qquad \sum_i \|r_i\|^2_{L^2(dx)} = d_n. \tag{2.26}$$

We give a general interpolation estimate before using the sublevel hypothesis. For real data $|a_i| \leq 1$, put

$$p = \sum_i a_i \ell_i, \qquad F_a = \sum_i \lambda_i a_i K_i, \qquad e = p - F_a = \sum_i a_i r_i.$$

Cauchy–Schwarz in this finite sum gives $\|e\|_{L^2(dx)} \leq \sqrt{n d_n}$. The interpolation equations $p(x_i) = a_i$ also give the exact identity

$$\langle p, F_a \rangle_{L^2(dx)} = \sum_i \lambda_i a_i p(x_i) = \sum_i \lambda_i a_i^2 \leq L_X.$$

Thus $u = \|p\|_{L^2(dx)}$ satisfies

$$u^2 = \langle p, F_a \rangle + \langle p, e \rangle \leq L_X + \sqrt{n d_n}\, u,$$

and solving this quadratic yields

$$\|p\|_{L^2(dx)} \leq \frac{\sqrt{n d_n} + \sqrt{n d_n + 4 L_X}}{2}. \tag{2.27}$$

Substitution of (2.25), followed by homogeneity in the data, proves (2.23).

The normalized nodal polynomial has leading coefficient 2^{n-1}, which is also the leading coefficient of T_n. Consequently

$$q_n = T_n - \sum_i T_n(x_i) \ell_i.$$

Since $|T_n(x_i)| \leq 1$ and $\|T_n\|_{L^2(dx)} \leq \sqrt{2}$, (2.23) proves $\|q_n\|_{L^2(dx)} \leq B$.

2.2.2 The circle logarithm

Define

$$R_n(z) = 2z^n q_n\left(\frac{z+z^{-1}}{2}\right) = \prod_i (1 - 2x_i z + z^2).$$

Each root pair is $e^{\pm i\theta_i}$, where $\theta_i = \arccos x_i$; an endpoint gives a double root. Define

$$v_n(\theta) = \log|R_n(e^{i\theta})| = \log|2q_n(\cos\theta)|.$$

Its finitely many logarithmic singularities are integrable. With $dm = d\theta/(2\pi)$ and $\widehat{v}_n(k) = \int v_n(\theta)e^{-ik\theta}\,dm$, the root factors give

$$\widehat{v}_n(0) = 0, \qquad \widehat{v}_n(k) = -\frac{1}{k}\sum_i T_k(x_i), \qquad k \geq 1. \tag{2.28}$$

Indeed, for $0 \leq r < 1$,

$$\log|1 - re^{it}| = -\sum_{k\geq 1}\frac{r^k}{k}\cos(kt).$$

The series may first be integrated for fixed $r < 1$. As r increases to 1, the logarithms converge in L^1 on the circle: near $t = 0$ their negative parts are bounded by a constant plus $|\log|t||$, since $|1 - re^{it}|^2 = (1-r)^2 + 2r(1-\cos t)$ for $r \geq 1/2$; their positive parts are uniformly bounded. Translation gives the same conclusion at any root. Each conjugate pair therefore contributes $-\cos(k\theta_i)/k$ to the positive Fourier coefficient. Multiplicity treats the endpoint pairs in exactly the same way.

For an even function let $\mathbb{E}v = \pi^{-1}\int_0^\pi v(\theta)\,d\theta$. Writing $v_n^+ = \max(v_n, 0)$ gives the pointwise estimate

$$v_n^+(\theta) \leq \frac{1}{2}\log\{1 + 4q_n(\cos\theta)^2\}.$$

Apply Jensen's inequality to $\{1 + 4q_n(\cos\theta)^2\}\sin\theta$. Its logarithm is integrable, including at the endpoints, and $\mathbb{E}\log\sin\theta = -\log 2$. Hence

$$\begin{aligned} 2\mathbb{E}v_n^+ &\leq \mathbb{E}\log\left[\{1 + 4q_n(\cos\theta)^2\}\sin\theta\right] - \mathbb{E}\log\sin\theta \\ &\leq \log\left\{\frac{1}{\pi}\int_{-1}^{1}(1 + 4q_n(x)^2)\,dx\right\} + \log 2 \\ &= \log\left(\frac{4 + 8\|q_n\|^2_{L^2(dx)}}{\pi}\right) \leq C. \end{aligned}$$

For completeness, the log-sine integral follows by applying $\sin(2\theta) = 2\sin\theta\cos\theta$ on $(0, \pi/2)$ and substituting $\theta \mapsto \pi/2 - \theta$ in the cosine term. By (2.28), $\mathbb{E}v_n = 0$, so $\mathbb{E}|v_n| = 2\mathbb{E}v_n^+ \leq C$. It follows that

$$\left|\sum_i T_k(x_i)\right| = k|\widehat{v}_n(k)| \leq k\mathbb{E}|v_n| \leq Ck,$$

which completes (2.24).

2.3 A reciprocal derivative bound

Let $n \geq 2$ and $I_n \leq 2$. With the constants in (2.22), put

$$J = \frac{(2 + \sqrt{D})^2}{2}. \tag{2.29}$$

We prove

$$\sum_i \frac{1}{q_n'(x_i)^2} \le \frac{J}{n}. \tag{2.30}$$

Let κ_m be the leading coefficient of $\psi_m = \sqrt{m+1/2}\,L_m$, and set $c_i = \langle \ell_i, \psi_{n-1}\rangle_{L^2(dx)}$. Degree orthogonality and comparison of leading coefficients give

$$c_i = \frac{1}{\kappa_{n-1} Q_n'(x_i)}, \qquad Q_n(x) = \prod_j (x - x_j). \tag{2.31}$$

The last identity in (2.9) implies

$$\lambda_n(x)\psi_{n-1}(x)^2 \le \frac{2n-1}{n^2} \le \frac{2}{n}, \qquad -1 \le x \le 1.$$

This includes the endpoints by the polynomial identity there. Project the residuals from (2.26) onto ψ_{n-1}. Then

$$\sum_i |c_i - \lambda_i \psi_{n-1}(x_i)|^2 \le d_n, \qquad \sum_i \lambda_i^2 \psi_{n-1}(x_i)^2 \le \frac{2n-1}{n^2} \sum_i \lambda_i \le \frac{4}{n}.$$

The triangle inequality for this finite coefficient vector and (2.25) yield

$$\sum_i c_i^2 \le \left(\sqrt{d_n} + \frac{2}{\sqrt{n}}\right)^2 \le \frac{(\sqrt{D}+2)^2}{n}. \tag{2.32}$$

The exact leading coefficient from (2.4) satisfies

$$v_m := \frac{\kappa_m^2}{4^m} = (m+\tfrac{1}{2})\frac{\binom{2m}{m}^2}{16^m}, \qquad v_0 = \frac{1}{2}, \qquad \frac{v_{m+1}}{v_m} = 1 - \frac{1}{4(m+1)^2} < 1.$$

Thus $v_m \le 1/2$ for all $m \ge 0$. Since $q_n = 2^{n-1} Q_n$, equation (2.31) gives

$$\sum_i \frac{1}{q_n'(x_i)^2} = \frac{\kappa_{n-1}^2}{4^{n-1}} \sum_i c_i^2 \le \frac{(\sqrt{D}+2)^2}{2n}.$$

This proves (2.30) with ordinary dx normalization throughout.

3 Analytic limits and the interpolation phase

3.1 Analytic compactness of sublevel configurations

We now fix a sequence of admissible configurations with $I_n \le 2$ and $n \to \infty$; all subsequences below refer to this sequence. The preceding estimates give absolute constants

$$D = 49 + 8\pi, \quad H = \frac{\sqrt{D} + \sqrt{D+8}}{2}, \quad B = \sqrt{2} + H, \quad J = \frac{(2+\sqrt{D})^2}{2}$$

such that, with the ordinary Legendre kernel K_n and $\lambda_n = 1/K_n(\cdot,\cdot)$ on the diagonal,

$$\lambda_n(x) \ge \frac{\pi\sqrt{1-x^2}}{n} - \frac{8\pi}{n^2}, \qquad \sum_i \sqrt{1-x_i^2} \ge \frac{2n-49}{\pi}, \tag{3.1}$$

$$d_n = I_n - \sum_i \lambda_n(x_i) = \sum_i \|\ell_i - \lambda_n(x_i) K_n(\cdot, x_i)\|_{L^2(dx)}^2, \qquad 0 \le n d_n \le D. \tag{3.2}$$

The degree-n nodal polynomial is normalized by

$$q_n(x) = 2^{n-1}\prod_i (x - x_i).$$

The preceding moment and top-degree estimates state

$$\|q_n\|_{L^2(dx)} \le B, \qquad \left|\sum_i T_k(x_i)\right| \le Ck \quad (k \ge 1), \qquad \sum_i \frac{1}{q_n'(x_i)^2} \le \frac{J}{n}, \quad C = \log\frac{4+8B^2}{\pi}. \tag{3.3}$$

Only constants independent of the node array will be used.

The moments encode the nodal polynomial in the disk. Put

$$R_n(z) = 2z^n q_n\left(\frac{z+z^{-1}}{2}\right) = \prod_i (1 - 2x_i z + z^2), \qquad P(z) = 1 - z^2.$$

The apparent singularity in the definition of R_n at zero is removable, and $R_n(0) = 1$. It has no zeros in $\mathbb{D}$, even when an original node is an endpoint. Choose its analytic logarithm to vanish at zero. If $S_{k,n} = \sum_i T_k(x_i)$, then

$$\log R_n(z) = -2\sum_{k\ge 1} \frac{S_{k,n}}{k} z^k.$$

Each real coefficient is bounded by $2C$. Successive coefficient extraction and a diagonal subsequence therefore give a real-coefficient analytic limit L with $L(0) = 0$. The geometric tail bound $2C\sum_{k>M} r^k$, on $|z| \le r < 1$, proves local uniform convergence. Thus, along that subsequence,

$$\log R_n \longrightarrow L, \qquad R_n \longrightarrow F = e^L \quad \text{locally uniformly in } \mathbb{D}, \qquad F(0) = 1. \tag{3.4}$$

In particular F is zero-free and has real Taylor coefficients. Local analytic convergence alone makes no boundary-trace claim.

Let $\theta_i = \arccos x_i$. The equal-atom angular probabilities

$$\nu_n = \frac{1}{n}\sum_i \delta_{\theta_i}$$

converge weakly to $\mathrm{d}\theta/\pi$ on $[0,\pi]$: their fixed nonconstant cosine moments are $S_{k,n}/n \to 0$, and cosine polynomials are uniformly dense in continuous functions. For example, this density follows from polynomial approximation after the homeomorphism $\theta \mapsto \cos\theta$; see [9]. If the angles are sorted increasingly and $Q_{i,n} = ((i-1)\pi/n, i\pi/n)$, continuity and strict monotonicity of the limiting distribution give

$$\max_i \sup_{s\in Q_{i,n}} |s - \theta_i| \longrightarrow 0. \tag{3.5}$$

Indeed a fixed positive quantile error would, by compactness, contradict convergence of the distribution functions to θ/π.

Set $v_n(s) = e^{2in\theta_i}$ on $Q_{i,n}$. These functions have modulus one, hence a further subsequence converges weakly in $L^2(0,\pi)$ to a function f with $|f| \le 1$ almost everywhere. One may construct this subsequence by taking diagonal limits of the coefficients in a countable orthonormal basis and representing the resulting bounded linear functional. The modulus bound follows by testing on measurable sets, or equivalently by weak closedness of the convex set $\{v : |v| \le 1\}$. Equation (3.5) identifies the corresponding continuous-test phase measures:

$$\frac{1}{n}\sum_i e^{2in\theta_i}\delta_{\theta_i} \rightharpoonup f(\theta)\frac{\mathrm{d}\theta}{\pi}. \tag{3.6}$$

The next subsection proves $|f| = 1$; it is not an assumption here.

On the full circle use normalized measure $dm = \mathrm{d}\theta/(2\pi)$ and the root probability

$$\mu_n = \frac{1}{2n} \sum_{\zeta : R_n(\zeta)=0} m_\zeta \delta_\zeta.$$

An interior node contributes two simple conjugate roots; an endpoint contributes one double root of mass $1/n$. Angular convergence shows $\mu_n \rightharpoonup dm$. Extend f by conjugate reflection to the lower semicircle. The full-circle phase measure has this same density, and restriction to a fixed bulk arc is legitimate because the limiting unphased measure has no atoms.

We use $H^2 = H^2(\mathbb{D})$ with norm $\|V\|_{H^2}^2 = \sum_{k\geq 0} |\widehat{V}(k)|^2$. Boundary traces, denoted by a star, are taken almost everywhere with respect to dm. The standard radial trace and L^2 convergence statements used below are those of [7, Theorem 5.8 and Corollary 5.26]. In particular a trace will only be used after its H^2 membership has been established.

3.2 The modulus of the limiting phase

Consider a sequence of distinct n-node sets, $n \geq 2$, with $I_n \leq 2$. Pass to a subsequence with the analytic limit $R_n \to F$ and the weak phase limit described above. Thus, with $\theta_i = \arccos x_i$,

$$\frac{1}{n}\sum_i \delta_{\theta_i} \rightharpoonup \frac{d\theta}{\pi}, \qquad \frac{1}{n}\sum_i e^{2in\theta_i}\delta_{\theta_i} \rightharpoonup f(\theta)\frac{d\theta}{\pi}, \qquad |f| \leq 1. \tag{3.7}$$

We prove that

$$|f(\theta)| = 1 \quad \text{for almost every } \theta \in (0,\pi). \tag{3.8}$$

The proof uses the actual Legendre kernel Gram matrix and retains its reflected term.

3.2.1 The Gram matrix on a fixed interior interval

Let U be the square matrix with rows $(\psi_0(x_i), \dots, \psi_{n-1}(x_i))$, and set

$$K = UU^*, \qquad \Lambda = \mathrm{diag}(\lambda_i), \qquad \mathcal{B} = \Lambda^{1/2} K \Lambda^{1/2}.$$

This positive definite matrix has diagonal one. The columns of U^{-1} are the cardinal coefficient vectors in the orthonormal Legendre basis. Consequently

$$I_n = \mathrm{tr}(\Lambda\mathcal{B}^{-1}), \qquad d_n = \sum_i \lambda_i\{(\mathcal{B}^{-1})_{ii} - 1\}. \tag{3.9}$$

Every bracket is nonnegative by a scalar Schur complement.

Fix $0 < \alpha < \pi/2$ and $J_\alpha = [\alpha, \pi - \alpha]$. Let $\mathcal{B}_J$ be the principal block indexed by the nodes whose angles belong to J_α, and let its size be $m_{n,\alpha}$. By (2.3), for all sufficiently large n,

$$\lambda_i \geq \frac{c_\alpha}{n} \quad (\theta_i \in J_\alpha), \qquad c_\alpha = \frac{\pi \sin\alpha}{2}.$$

The Schur inverse inequality $(\mathcal{B}^{-1})_{JJ} \succeq \mathcal{B}_J^{-1}$, applied in (3.9), gives

$$\mathrm{tr}\,\mathcal{B}_J^{-1} - m_{n,\alpha} \leq T_\alpha := \frac{D}{c_\alpha}.$$

Since $\mathrm{tr}\,\mathcal{B}_J = m_{n,\alpha}$, its positive eigenvalues b satisfy

$$\sum_b \frac{(b-1)^2}{b} = \mathrm{tr}\,\mathcal{B}_J + \mathrm{tr}\,\mathcal{B}_J^{-1} - 2m_{n,\alpha} \leq T_\alpha.$$

In particular, with $L_\alpha = T_\alpha + 2$,

$$L_\alpha^{-1}\mathrm{Id} \preceq \mathcal{B}_J \preceq L_\alpha \mathrm{Id}, \qquad \|\mathcal{B}_J - \mathrm{Id}\|_F^2 \le L_\alpha T_\alpha. \tag{3.10}$$

Indeed $b + b^{-1} \le T_\alpha + 2$ for each eigenvalue, and multiplication of $(b-1)^2/b$ by $b \le L_\alpha$ gives the Frobenius bound. Only this interior block has been estimated; the weights at all other nodes remain in the original expression (3.9).

3.2.2 The full kernel away from the diagonal

For fixed $\varepsilon > 0$, uniformly for $\theta, \varphi \in J_\alpha$ with $|\theta - \varphi| \ge \varepsilon$,

$$\frac{K_n(\cos\theta, \cos\varphi)}{\sqrt{K_n(\cos\theta,\cos\theta)K_n(\cos\varphi,\cos\varphi)}} = \frac{1}{2n}\left\{\frac{\sin[n(\theta-\varphi)]}{\sin[(\theta-\varphi)/2]} - \frac{\cos[n(\theta+\varphi)]}{\sin[(\theta+\varphi)/2]}\right\} + O_{\alpha,\varepsilon}(n^{-2}). \tag{3.11}$$

Also, uniformly on J_α,

$$n\lambda_n(\cos\theta) = \pi\sin\theta + O_\alpha(n^{-1}). \tag{3.12}$$

Here is a derivation that does not differentiate an asymptotic error. The compact-interior Darboux formula [6, Eq. 18.15.4_5], specialized to Legendre polynomials, gives

$$g_n(\theta) := \sqrt{n}\, L_n(\cos\theta) = a(\theta)\cos\chi_n + O_\alpha(n^{-1}),$$

$$h_n(\theta) := \sqrt{n}\, L_{n-1}(\cos\theta) = a(\theta)\cos(\chi_n - \theta) + O_\alpha(n^{-1}),$$

where

$$a(\theta) = \sqrt{\frac{2}{\pi\sin\theta}}, \quad \chi_n = (n+\tfrac{1}{2})\theta - \tfrac{\pi}{4}.$$

The exact lowering identity gives

$$g_n' = n\frac{\cos\theta\, g_n - h_n}{\sin\theta} = -na(\theta)\sin\chi_n + O_\alpha(1).$$

Using (2.9) with L_n now proves $K_n(\cos\theta,\cos\theta) = n/(\pi\sin\theta) + O_\alpha(1)$, and hence (3.12).

For distinct arguments (2.8) becomes

$$K_n(\cos\theta,\cos\varphi) = \frac{1}{2}\frac{g_n(\theta)h_n(\varphi) - h_n(\theta)g_n(\varphi)}{\cos\theta - \cos\varphi}.$$

The leading trigonometric numerator is

$$a(\theta)a(\varphi)\left\{-\sin[n(\theta-\varphi)]\sin\frac{\theta+\varphi}{2} + \cos[n(\theta+\varphi)]\sin\frac{\theta-\varphi}{2}\right\}.$$

On the fixed region under consideration the numerator error is $O(n^{-1})$ and $|\cos\theta - \cos\varphi|$ is bounded away from zero. Thus the unnormalized kernel equals

$$\frac{1}{2\pi\sqrt{\sin\theta\sin\varphi}}\left\{\frac{\sin[n(\theta-\varphi)]}{\sin[(\theta-\varphi)/2]} - \frac{\cos[n(\theta+\varphi)]}{\sin[(\theta+\varphi)/2]}\right\} + O_{\alpha,\varepsilon}(n^{-1}).$$

Its normalizing factor is $n/(\pi\sqrt{\sin\theta\sin\varphi})\{1 + O_\alpha(n^{-1})\}$. The expression in braces is bounded on this fixed region, proving (3.11). No estimate uniform down to the diagonal or to an endpoint has been used.

3.2.3 Equal cells and the weak kernel limit

Relabel the angles increasingly and put

$$\ell_n = \frac{\pi}{n}, \quad Q_{i,n} = ((i-1)\ell_n, i\ell_n), \qquad t_n(s) = \theta_i, \quad v_n(s) = e^{2in\theta_i} \quad (s \in Q_{i,n}).$$

The distribution functions in (3.7) converge uniformly: a fixed fine partition bounds their differences at all points by monotonicity and the continuity of the limiting distribution function. The ordered atoms therefore satisfy

$$\epsilon_n := \max_i \sup_{s \in Q_{i,n}} |s - \theta_i| \longrightarrow 0. \tag{3.13}$$

This is an unscaled transport estimate; the phases retain their actual node arguments θ_i. Uniform continuity of continuous tests, followed by their density in $L^2(0,\pi)$, shows that $v_n \rightharpoonup f$ in $L^2(0,\pi)$. Moreover

$$\overline{v_n(s)} v_n(t) \rightharpoonup \overline{f(s)} f(t) \quad \text{in } L^2((0,\pi)^2). \tag{3.14}$$

For a separated test this follows by factoring its integral into two one-variable weak pairings. Finite sums of separated tests are dense in L^2 of the square, and the product functions have a common L^2 bound, so the conclusion extends to every L^2 test.

Let J_n be the union of the cells whose actual angles lie in J_α. Its endpoints tend to those of J_α by (3.13). On that block let $Z_n = \operatorname{diag}(e^{in\theta_i})$, and put

$$\widehat{\mathcal{C}}_n = Z_n^* \mathcal{B}_J Z_n - \mathrm{Id}.$$

Its diagonal is exactly zero. Map the block coordinate vector e_i isometrically to $\ell_n^{-1/2} 1_{Q_{i,n}}$ in $\mathcal{H} = L^2((0,\pi);\mathbb{C})$, and denote the isometry by E_n. The operator $\mathcal{C}_n = E_n \widehat{\mathcal{C}}_n E_n^*$ has integral kernel $\widehat{\mathcal{C}}_{n,ij}/\ell_n$ on $Q_{i,n} \times Q_{j,n}$, and zero on the diagonal cells and outside J_n^2. In particular,

$$\|\mathcal{C}_n\|_{\mathfrak{S}_2} = \|\widehat{\mathcal{C}}_n\|_F \le \sqrt{L_\alpha T_\alpha}, \qquad L_\alpha^{-1}\mathrm{Id} \preceq \mathrm{Id} + \mathcal{C}_n \preceq L_\alpha \mathrm{Id}. \tag{3.15}$$

On the orthogonal complement of the active cell space, $\mathrm{Id} + \mathcal{C}_n$ is the identity, so these are exact bounds on the fixed Hilbert space $\mathcal{H}$.

Away from each fixed diagonal strip, dephasing (3.11) and dividing by ℓ_n gives the cell kernel

$$\begin{aligned} &\frac{1 - \overline{v_n(s)} v_n(t)}{4\pi i \sin[(t_n(s) - t_n(t))/2]} \\ &\qquad - \frac{v_n(t) + \overline{v_n(s)}}{4\pi \sin[(t_n(s) + t_n(t))/2]} + O_{\alpha,\varepsilon}(n^{-1}) \end{aligned} \tag{3.16}$$

on the active cells. The two dephasing identities that fix its signs are

$$e^{-in(\theta-\varphi)} \sin[n(\theta - \varphi)] = \frac{1 - e^{-2in\theta} e^{2in\varphi}}{2i},$$

$$e^{-in(\theta-\varphi)} \cos[n(\theta + \varphi)] = \frac{e^{2in\varphi} + e^{-2in\theta}}{2}.$$

The denominators converge uniformly on each such fixed region by (3.13); the active-cell indicators converge in measure to 1_{J_α}. By (3.14) the off-diagonal weak limit is therefore $1_{J_\alpha}(s) 1_{J_\alpha}(t) c_f(s,t)$, where

$$c_f(s,t) = \frac{1 - \overline{f(s)} f(t)}{4\pi i \sin((s-t)/2)} - \frac{f(t) + \overline{f(s)}}{4\pi \sin((s+t)/2)}. \tag{3.17}$$

The bound (3.15) supplies weakly convergent L^2 subsequences of these kernels. Every limit has the preceding off-diagonal expression and is supported on J_α^2. The diagonal has planar measure zero, so this identifies a unique L^2 kernel. Equivalently, a uniformly L^2 bounded remainder on a shrinking diagonal strip has vanishing pairing with each fixed L^2 test: Cauchy–Schwarz bounds it by a common constant times the test's L^2 norm on that strip, which tends to zero. Thus

$$\mathcal{C}_n \rightharpoonup \mathcal{C}_f^{J_\alpha} \quad \text{weakly in } \mathfrak{S}_2(\mathcal{H}),$$

where the limit has kernel $1_{J_\alpha}(s)1_{J_\alpha}(t)c_f(s,t)$. This statement concerns weak kernel convergence and does not discard a possible concentration of the squared kernel norm near the diagonal.

3.2.4 The diagonal singularity forces unit modulus

The reflected term of (3.17) is bounded on J_α^2. Its first term is consequently in $L^2(J_\alpha^2)$. Since $|f| \le 1$,

$$|1 - \overline{f(s)}f(t)| \ge 1 - |f(s)|.$$

If $|f(s)| < 1$ on a set of positive measure, Tonelli's theorem and

$$\int_{J_\alpha} \frac{dt}{\sin^2((s-t)/2)} = +\infty \quad \text{for every interior } s \in J_\alpha$$

contradict this L^2 property. It follows that $|f| = 1$ almost everywhere on J_α. Letting α range through a sequence decreasing to zero proves (3.8). In addition,

$$\iint_{J_\alpha^2} \frac{|f(s) - f(t)|^2}{\sin^2((s-t)/2)}\, ds\, dt < \infty,$$

since $|1 - \overline{f(s)}f(t)| = |f(s) - f(t)|$ there. Weak convergence of v_n and equality of the limiting norms also give $v_n \to f$ strongly in $L^2(0,\pi)$. Finally self-adjointness and the order bounds in (3.15) pass to the weak limit by testing against rank-one kernels, so

$$L_\alpha^{-1}\mathrm{Id} \preceq \mathrm{Id} + \mathcal{C}_f^{J_\alpha} \preceq L_\alpha \mathrm{Id}.$$

3.3 Reciprocal sampling and Hardy limits

Continue along the subsequence of Section 3.2. On the full circle let

$$\mu_n = \frac{1}{2n} \sum_{R_n(\zeta)=0} m_\zeta \delta_\zeta, \qquad \eta_n = \zeta^{2n}\mu_n.$$

The angular equidistribution gives $\mu_n \rightharpoonup dm$. Conjugate reflection of (3.7) gives $\eta_n \rightharpoonup f\, dm$ on the full circle, where $f(e^{-i\theta}) = \overline{f(e^{i\theta})}$ and $|f| = 1$ almost everywhere by (3.8). These statements extend across the two real endpoints: $|\eta_n| \le \mu_n$ bounds tests in shrinking endpoint neighborhoods, while their limiting unphased mass tends to zero. Each endpoint atom has mass $1/n$.

We shall prove

$$A = \sqrt{P}\, F \in H^2(\mathbb{D}), \qquad G = \frac{P}{F} \in H^2(\mathbb{D}), \qquad \|A\|_{H^2}^2 \le \frac{8B^2}{\pi}, \quad \|G\|_{H^2}^2 \le 4J, \tag{3.18}$$

and identify the phase as

$$f = -\frac{F^*}{\overline{F^*}} \quad \text{almost everywhere on the circle.} \tag{3.19}$$

Here $P(z) = 1 - z^2$, its square root equals one at zero, and F^* is the local L^2 boundary trace away from ± 1 constructed below.

3.3.1 An exact rational identity, including double roots

At the atom associated with x_i, define

$$g_n(\zeta) = \frac{2n\,\zeta^{1-n}}{q_n'(x_i)}.$$

The same formula applies to a double root $\zeta = \pm 1$. Every original node has total circle mass $1/n$. Consequently

$$\int |g_n|^2\, d\mu_n = 4n \sum_i \frac{1}{q_n'(x_i)^2} \le 4J. \tag{3.20}$$

For $n \ge 2$ the following identity holds throughout the disk:

$$\int \frac{g_n(\zeta)}{1-\overline{\zeta}z}\, d\mu_n(\zeta) = \frac{P(z)}{R_n(z)}. \tag{3.21}$$

More generally, for every integer $k \ge 0$ with $k+2 \le 2n-1$,

$$\int \frac{\zeta^k g_n(\zeta)}{1-\overline{\zeta}z}\, d\mu_n(\zeta) = \frac{z^k P(z)}{R_n(z)}. \tag{3.22}$$

We verify these as identities of proper rational functions. At an interior root ζ, differentiation of $R_n(z) = 2z^n q_n((z+z^{-1})/2)$ gives

$$R_n'(\zeta) = -\zeta^{n-2} P(\zeta) q_n'(x_i).$$

The partial-fraction coefficient of $z^k P/R_n$ at its simple pole ζ is therefore $-\zeta^{k+2-n}/q_n'(x_i)$. Since $1/(1-\overline{\zeta}z) = -\zeta/(z-\zeta)$ and the atom has mass $1/(2n)$, its coefficient on the left side of (3.22) is exactly the same.

At an endpoint root $\zeta = \pm 1$, the exact relation

$$\frac{z+z^{-1}}{2} - \zeta = \frac{(z-\zeta)^2}{2z}$$

shows that $R_n(z) = (z-\zeta)^2 T(z)$ with $T(\zeta) = \zeta^{n-1} q_n'(x_i)$. The zero of P reduces this double pole to a simple pole with coefficient

$$\frac{\zeta^k P'(\zeta)}{T(\zeta)} = -\frac{2\zeta^{k+2-n}}{q_n'(x_i)}.$$

The atom now has mass $1/n$, so its coefficient in the integral is again the same. All poles have been accounted for, and both sides are proper because $k+2 < 2n$. Thus their complete finite partial fractions coincide, proving (3.22). Its case $k=0$ is (3.21). At $z=0$ its cases $k \ge 1$ give

$$\int \zeta^k g_n(\zeta)\, d\mu_n(\zeta) = 0 \quad \text{for every fixed } k \ge 1 \text{ and all sufficiently large } n. \tag{3.23}$$

3.3.2 All Fourier coefficients of the reciprocal limit

Write

$$\frac{P}{R_n} = \sum_{m \ge 0} a_{m,n} z^m, \qquad \frac{P}{F} = \sum_{m \ge 0} a_m z^m.$$

Local analytic convergence gives $a_{m,n} \to a_m$ for every fixed m, while (3.21) gives

$$a_{m,n} = \int g_n(\zeta) \overline{\zeta}^{\,m}\, d\mu_n(\zeta).$$

For fixed M, use the continuous polynomial $h_M(\zeta) = \sum_{m=0}^{M} \overline{a_m}\,\overline{\zeta}^{\,m}$. Cauchy–Schwarz and (3.20) imply

$$\sum_{m=0}^{M} |a_m|^2 = \lim_n \int h_M g_n \, d\mu_n$$
$$\leq 2\sqrt{J} \lim_n \left(\int |h_M|^2 \, d\mu_n \right)^{1/2} = 2\sqrt{J} \left(\sum_{m=0}^{M} |a_m|^2 \right)^{1/2}.$$

Every partial square sum is at most $4J$. This proves $G = P/F \in H^2$ and the bound for G in (3.18). Its boundary trace G^* has the same nonnegative Fourier coefficients a_m and zero negative Fourier coefficients. By the preceding identities and (3.23), $g_n\mu_n$ converges to G^*dm on every trigonometric polynomial. Its total variation is at most $2\sqrt{J}$, and $\|G^*\|_{L^1(dm)} \leq 2\sqrt{J}$. Uniform trigonometric approximation of continuous functions therefore proves

$$g_n\mu_n \rightharpoonup G^*dm. \tag{3.24}$$

No additional subsequence is needed for this limit.

3.3.3 Boundary traces and nonvanishing

The change $x = \cos\theta$ gives the exact norm identity

$$\|\sqrt{P}\, R_n\|_{H^2}^2 = \int |1 - e^{2i\theta}|\, |2q_n(\cos\theta)|^2 \, dm = \frac{8}{\pi} \|q_n\|_{L^2(dx)}^2 \leq \frac{8B^2}{\pi}. \tag{3.25}$$

Coefficient convergence and finite partial square sums yield $A = \sqrt{P}\, F \in H^2$ and its stated bound. The L^2 boundary trace of A, divided on each compact arc away from ± 1 by the smooth nonzero trace of $\sqrt{P}$, defines F^*. The radial functions $F(r\zeta)$ converge to F^* in L^2 on each such arc; the radial functions of G converge in L^2 on the whole circle. The disk identity $FG = P$ and Cauchy–Schwarz therefore imply

$$F^*G^* = P \quad \text{almost everywhere away from } \pm 1. \tag{3.26}$$

Indeed, on a compact arc the product differences tend to zero in L^1. Since P is nonzero there, both traces are nonzero almost everywhere.

3.3.4 Passing a phase-times-value product to the limit

Let $u_n(\zeta) = \zeta^{2n}$. For every fixed continuous circle function φ, the two measure limits and $|f| = 1$ give

$$\lim_n \int |u_n - \varphi|^2 \, d\mu_n = \int |f - \varphi|^2 \, dm. \tag{3.27}$$

This follows by expanding the square; its constant term is one on both sides, while the cross term is determined by $\eta_n \rightharpoonup f\, dm$. The definition of g_n and the reality of $q_n'(x_i)$ give the exact sampled identity

$$u_n(\zeta) g_n(\zeta) = \zeta^2 \overline{g_n(\zeta)}. \tag{3.28}$$

At $\zeta = \pm 1$ the sampled value is real and both powers in this identity equal one, so it includes endpoint atoms.

For a continuous test h, choose continuous φ approximating f in $L^2(dm)$. Equations (3.20) and (3.27) give

$$\limsup_n \left| \int h(u_n - \varphi) g_n \, d\mu_n \right| \leq 2\sqrt{J}\, \|h\|_\infty \|f - \varphi\|_{L^2(dm)}.$$

For fixed φ, (3.24) passes $\int h\varphi g_n\,d\mu_n$ to $\int h\varphi G^*\,dm$. Letting $\varphi \to f$ in L^2 and using $G^* \in L^2$ now shows

$$u_n g_n \mu_n \rightharpoonup f G^* dm.$$

Conjugating (3.24) passes the right side of (3.28) to $\zeta^2\overline{G^*}dm$. Consequently $fG^* = \zeta^2\overline{G^*}$ almost everywhere. Finally $\overline{P(\zeta)} = -\zeta^{-2}P(\zeta)$ on the circle, and (3.26) gives

$$f = \zeta^2\frac{\overline{G^*}}{G^*} = -\frac{F^*}{\overline{F^*}},$$

which proves (3.19). The argument uses continuous approximants to the limiting phase and never evaluates an arbitrary boundary representative at the moving nodes.

4 First-order corrections to the energy

4.1 The scalar angular deficit

Along the selected sequence with $I_n \le 2$ and $R_n \to F$ locally uniformly, define

$$a_n = 2n - \pi\sum_i \sqrt{1-x_i^2}.$$

We prove

$$a_n \longrightarrow A_{\mathrm{scalar}}(F), \qquad A_{\mathrm{scalar}}(F) = -\frac{1}{2}\int_0^1 (1+r^{-2})\log[F(r)F(-r)]\,dr. \tag{4.1}$$

The real-axis values $F(\pm r)$ are positive for $0 \le r < 1$; the logarithm in this expression is real, and the improper integral is finite.

4.1.1 Two-sided bounds on real radii

From (3.21) and (3.20), for $|z| < 1$,

$$\left|\frac{P(z)}{R_n(z)}\right| \le \frac{2\sqrt{J}}{1-|z|}.$$

Every factor $1 \mp 2x_i r + r^2$ of $R_n(\pm r)$ is positive for $0 \le r < 1$. Therefore

$$R_n(\pm r) \ge \frac{(1-r^2)(1-r)}{2\sqrt{J}}. \tag{4.2}$$

Put $K_A = \sqrt{8/\pi}\,B$. The point-evaluation estimate for an H^2 Taylor series, applied to (3.25), gives

$$|\sqrt{P(\pm r)}\,R_n(\pm r)| \le \frac{K_A}{\sqrt{1-r^2}}.$$

Since $\sqrt{P(\pm r)} = \sqrt{1-r^2} > 0$,

$$R_n(\pm r) \le \frac{K_A}{1-r^2}. \tag{4.3}$$

It follows that, with an absolute constant C_0,

$$|\log R_n(\pm r)| \le C_0 + 2|\log(1-r)|, \qquad \tfrac{1}{2} \le r < 1. \tag{4.4}$$

This is the real-radius bound needed for the improper integral.

4.1.2 An exact identity for a single node

For $-1 \le x \le 1$, put

$$D_x(r) = (1+r^2)^2 - 4x^2r^2, \qquad K(x) = -\frac{1}{2}\int_0^1 (1+r^{-2}) \log D_x(r)\, dr.$$

The logarithm is $O(r^2)$ at zero. At $x = \pm 1$ its only additional singularity is logarithmic at $r = 1$, so the displayed integral exists. We claim

$$K(x) = 2 - \pi\sqrt{1-x^2}. \tag{4.5}$$

At $x = 0$, direct integration gives

$$\int_0^1 \log(1+r^2)\, dr = \log 2 - 2 + \frac{\pi}{2}, \qquad \int_0^1 r^{-2} \log(1+r^2)\, dr = -\log 2 + \frac{\pi}{2},$$

and hence $K(0) = 2 - \pi$. For $|x| < 1$, differentiation under the integral is dominated on every compact x-subinterval and yields

$$K'(x) = 4x \int_0^1 \frac{1+r^2}{D_x(r)}\, dr.$$

With $t = (1-r^2)/(2r)$ and $s = \sqrt{1-x^2}$, the integral on the right is

$$\frac{1}{2}\int_0^\infty \frac{dt}{t^2+s^2} = \frac{\pi}{4s}.$$

Thus $K'(x) = \pi x/\sqrt{1-x^2}$, which proves (4.5) in $(-1, 1)$. As $|x|$ increases to one, the function

$$\frac{1}{2}(1+r^{-2}) \log \frac{D_0(r)}{D_x(r)}$$

is nonnegative and increases pointwise. Monotone convergence applied to $K(x) - K(0)$ gives $K(\pm 1) = 2$, proving the endpoint cases as well.

For a finite node set, $R_n(r)R_n(-r) = \prod_i D_{x_i}(r)$. Summing (4.5) therefore gives the exact formula

$$a_n = -\frac{1}{2}\int_0^1 (1+r^{-2}) \log[R_n(r)R_n(-r)]\, dr. \tag{4.6}$$

All endpoint terms remain in this finite sum of integrable functions.

4.1.3 The cancellation at zero and the limit

Let $S_{k,n} = \sum_i T_k(x_i)$. The logarithm normalized to vanish at zero is

$$\log R_n(z) = -2\sum_{k\ge 1} \frac{S_{k,n}}{k} z^k.$$

The bound $|S_{k,n}| \le Ck$ in (2.24) gives

$$|\log[R_n(r)R_n(-r)]| \le \frac{4Cr^2}{1-r^2}. \tag{4.7}$$

The odd powers cancel, so multiplication by $1 + r^{-2}$ leaves a uniform bound near zero. On $[1/2, 1)$, (4.4) supplies an integrable majorant of the form $C_1 + C_2|\log(1-r)|$ for the integrand of (4.6). Local convergence and the zero-free property of F imply $F(\pm r) > 0$ and convergence of the real logarithms for every $r < 1$. Dominated convergence in (4.6) now proves (4.1), including finiteness at both ends of its integral.

4.2 The first-order Christoffel contribution

For every fixed $0 < \varepsilon < \pi/2$, we shall prove, uniformly for $\varepsilon \le \theta \le \pi - \varepsilon$,

$$K_n(\cos\theta, \cos\theta) = \frac{n}{\pi \sin\theta} - \frac{\cos(2n\theta)}{2\pi \sin^2\theta} + O_\varepsilon(n^{-1}), \tag{4.8}$$

$$\lambda_n(\cos\theta) = \frac{\pi \sin\theta}{n} + \frac{\pi \cos(2n\theta)}{2n^2} + O_\varepsilon(n^{-3}). \tag{4.9}$$

We then use the global bound (2.3) to pass to a one-sided limit for the actual nodes, without extending either expansion to an endpoint.

4.2.1 Midpoint normalization

For a polynomial degree m, let

$$h = \sin\theta, \quad u_m(\theta) = \sqrt{h}\, L_m(\cos\theta), \quad L = m + \tfrac{1}{2}, \quad V(\theta) = \frac{1}{4\sin^2\theta}.$$

Direct substitution into (2.6) gives

$$u_m'' + (L^2 + V)u_m = 0. \tag{4.10}$$

We need its midpoint data to relative order L^{-2}. Set

$$C_r = 4^{-r}\binom{2r}{r}, \qquad A_L = \sqrt{\frac{2}{\pi L}}, \qquad \chi = L\theta - \frac{\pi}{4}.$$

The following refinement of the Wallis estimate is sufficient:

$$a_r := \sqrt{\pi(r + \tfrac{1}{4})}\, C_r = 1 + O(r^{-2}). \tag{4.11}$$

To prove it, the integrals used for (2.10) also give

$$J_{2r} = \frac{\pi}{2} C_r, \qquad J_{2r+1} = \frac{1}{(2r+1)C_r}, \qquad J_{2r-1} = \frac{1}{2rC_r}.$$

The strict inequalities $J_{2r+1} < J_{2r} < J_{2r-1}$ imply

$$\frac{1}{\pi(r + 1/2)} < C_r^2 < \frac{1}{\pi r},$$

so $a_r \to 1$. The exact recurrence for C_r gives

$$\frac{a_{r+1}^2}{a_r^2} = \frac{(r+1/2)^2(r+5/4)}{(r+1)^2(r+1/4)} = 1 + \frac{1}{16(r+1)^2(r+1/4)}.$$

Hence $\log(a_{r+1}/a_r) = O(r^{-3})$; summing from r to infinity, using the known limit, proves $\log a_r = O(r^{-2})$ and (4.11).

For even $m = 2r$, the exact midpoint values in (2.11) and $L = 2(r + 1/4)$ imply

$$u_m(\pi/2) = A_L \cos\chi(\pi/2)\{1 + O(L^{-2})\}, \qquad u_m'(\pi/2) = 0. \tag{4.12}$$

For odd $m = 2r + 1$, $u_m(\pi/2) = 0$, while

$$\frac{(2r+1)C_r}{A_L L} = a_r \frac{r + 1/2}{\sqrt{(r+1/4)(r+3/4)}} = 1 + O(r^{-2}).$$

Indeed the square of the last ratio after a_r is $1 + 1/[16(r + 1/4)(r + 3/4)]$. Since $x = \cos\theta$ reverses the derivative sign at the midpoint, this gives

$$u_m'(\pi/2) = -A_L L \sin\chi(\pi/2)\{1 + O(L^{-2})\}. \tag{4.13}$$

Only large m is relevant to the expansion, so these estimates require no separate convention at the finite small-degree prefix.

4.2.2 An expansion with a derivative remainder

Put $a(\theta) = \cot\theta/8$ and use the trial function

$$v_m(\theta) = A_L\left\{\cos\chi + \frac{a(\theta)}{L}\sin\chi\right\}.$$

Since $2a' + V = 0$, direct differentiation gives the residual

$$\rho_m := v_m'' + (L^2 + V)v_m = \frac{A_L}{L}(a'' + Va)\sin\chi = O_\varepsilon(A_L/L) \tag{4.14}$$

on $[\varepsilon, \pi - \varepsilon]$. Let $e = u_m - v_m$. Equations (4.12) and (4.13), together with $a(\pi/2) = 0$ and $a'(\pi/2) = -1/8$, give

$$e(\pi/2) = O(A_L/L^2), \qquad e'(\pi/2) = O(A_L/L).$$

The exact variation-of-constants formula for $e'' + L^2 e = -Ve - \rho_m$ is

$$\begin{aligned} e(\theta) = e(\pi/2)\cos[L(\theta - \pi/2)] + \frac{e'(\pi/2)}{L}\sin[L(\theta - \pi/2)] \\ - \frac{1}{L}\int_{\pi/2}^{\theta} \sin[L(\theta - t)]\{V(t)e(t) + \rho_m(t)\}\,dt. \end{aligned} \tag{4.15}$$

This oriented integral formula holds on either side of $\pi/2$ and can be checked by two differentiations and its initial values. The supremum of V and the interval length are bounded for fixed ε. Taking suprema in (4.15) therefore gives

$$\|e\|_\infty \le C_\varepsilon A_L L^{-2} + \frac{C_\varepsilon}{L}\|e\|_\infty.$$

Absorbing the last term for large L proves $\|e\|_\infty = O_\varepsilon(A_L L^{-2})$. Differentiating the exact formula (4.15), whose integrands are smooth, then gives $\|e'\|_\infty = O_\varepsilon(A_L L^{-1})$. Thus

$$u_m = A_L\{\cos\chi + (a/L)\sin\chi + E_m\}, \qquad \|E_m\|_\infty \le C_\varepsilon L^{-2}, \quad \|E_m'\|_\infty \le C_\varepsilon L^{-1}. \tag{4.16}$$

In particular, the derivative estimate follows from the differential equation, rather than from differentiating an unspecified asymptotic remainder.

4.2.3 The first-order diagonal kernel

Take $m = n - 1$, so $L = n - 1/2$, and put $c = \cos\chi$, $s = \sin\chi$, and $b = \cot\theta/2$. The last identity in (2.9) becomes

$$2K_n(\cos\theta, \cos\theta) = \frac{n^2u_m^2 + (u_m' - bu_m)^2}{h}. \tag{4.17}$$

Equation (4.16) gives, uniformly on the fixed interior interval,

$$\frac{u_m}{A_L} = c + \frac{a}{L}s + O_\varepsilon(L^{-2}), \qquad \frac{u_m' - bu_m}{A_L} = -Ls + (a - b)c + O_\varepsilon(L^{-1}).$$

After squaring and retaining every term that can have order L,

$$\begin{aligned} \frac{n^2u_m^2 + (u_m' - bu_m)^2}{A_L^2} &= n^2c^2 + L^2s^2 + 2\{n^2a/L - L(a - b)\}cs + O_\varepsilon(1) \\ &= L^2 + (L + \tfrac{1}{4})c^2 + 2Lbcs + O_\varepsilon(1). \end{aligned}$$

Here $n^2 - L^2 = L + 1/4$, and $n^2 a/L - L(a-b) = Lb + O_\varepsilon(1)$. Since $A_L^2 = 2/(\pi L)$, substitution into (4.17) yields

$$K_n(\cos\theta, \cos\theta) = \frac{n}{\pi h} + \frac{\frac{1}{2}\cos(2\chi) + b\sin(2\chi)}{\pi h} + O_\varepsilon(n^{-1}).$$

Now $2\chi = (2n-1)\theta - \pi/2$, so its correction numerator equals

$$\frac{1}{2}\sin[(2n-1)\theta] - \frac{1}{2}\cot\theta\cos[(2n-1)\theta] = -\frac{\cos(2n\theta)}{2\sin\theta}.$$

This proves (4.8), with its stated sign and frequency. Factoring the leading term gives

$$K_n(\cos\theta, \cos\theta) = \frac{n}{\pi h}\left\{1 - \frac{\cos(2n\theta)}{2nh} + O_\varepsilon(n^{-2})\right\}.$$

The bracket is bounded away from zero for large n; expanding its reciprocal with its uniform quadratic error proves (4.9). The exact midpoint formula provides a further check on the normalization:

$$K_n(0,0) = \frac{n^2}{2}C^2_{\lfloor n/2\rfloor} = \frac{n}{\pi} - \frac{(-1)^n}{2\pi} + O(n^{-1}),$$

by (4.11), in agreement with (4.8) at $\theta = \pi/2$.

4.2.4 A one-sided passage through the endpoint layers

Return to the actual node sequence and define

$$b_n = n\sum_i \left\{\lambda_n(x_i) - \frac{\pi\sin\theta_i}{n}\right\}, \qquad \beta_{i,n} = n^2\left\{\lambda_n(x_i) - \frac{\pi\sin\theta_i}{n}\right\}.$$

Thus $b_n = n^{-1}\sum_i \beta_{i,n}$. The global estimate (2.3) gives $\beta_{i,n} \geq -8\pi$ at every node, including an endpoint. We claim

$$\liminf_n b_n \geq \frac{1}{2}\int_0^\pi \operatorname{Re} f(\theta)\, d\theta. \tag{4.18}$$

For fixed $\varepsilon > 0$, let $J_\varepsilon = [\varepsilon, \pi - \varepsilon]$. The expansion (4.9) gives

$$\frac{1}{n}\sum_{\theta_i \in J_\varepsilon} \beta_{i,n} = \frac{\pi}{2n}\sum_{\theta_i \in J_\varepsilon} \cos(2n\theta_i) + O_\varepsilon(n^{-1}).$$

Restriction of the weak phase limit (3.7) to this interval is legitimate: its endpoints have zero mass for the limiting unphased measure, which dominates the phase measures. Hence the last expression converges to $\frac{1}{2}\int_\varepsilon^{\pi-\varepsilon} \operatorname{Re} f\, d\theta$. For the omitted nodes use their global lower bound alone:

$$\frac{1}{n}\sum_{\theta_i \notin J_\varepsilon} \beta_{i,n} \geq -8\pi\frac{\#\{i : \theta_i \notin J_\varepsilon\}}{n}.$$

Angular equidistribution makes the omitted fraction tend to $2\varepsilon/\pi$. Thus

$$\liminf_n b_n \geq \frac{1}{2}\int_\varepsilon^{\pi-\varepsilon} \operatorname{Re} f\, d\theta - 16\varepsilon.$$

Since $|f| \leq 1$, sending $\varepsilon \downarrow 0$ proves (4.18). The limits were taken first in n with ε fixed and then in ε. In particular no O_ε estimate was applied uniformly at an endpoint. An exact endpoint node has $\lambda_n(\pm 1) = 2/n^2$ and $\beta_{i,n} = 2$, so it is already included in this argument.

4.3 The coefficient retained by the original energy

For the sublevel sequence and the analytic factor F constructed above, the phase and reciprocal results give

$$A = \sqrt{P}\,F \in H^2, \qquad G = P/F \in H^2, \qquad AG = P^{3/2}, \qquad f = -F^*/\overline{F^*} \quad \text{a.e.}$$

All analytic powers of P equal one at zero. Define the real quantities

$$a_n = 2n - \pi \sum_i \sin\theta_i, \quad \Delta_n = n d_n, \quad b_n = n \sum_i \left(\lambda_n(x_i) - \frac{\pi \sin\theta_i}{n} \right), \quad c_n = n(2 - I_n).$$

By the definition of d_n,

$$c_n = a_n - \Delta_n - b_n. \tag{4.19}$$

The bounds (3.1)–(3.2) give

$$-8\pi \le a_n \le 49, \quad 0 \le \Delta_n \le D, \quad -8\pi \le b_n \le 49.$$

For the last upper bound use $c_n \ge 0$ and $\Delta_n \ge 0$ in (4.19).

The scalar and Christoffel estimates of the preceding sections yield

$$a_n \longrightarrow \mathcal{A}(F) = -\frac{1}{2} \int_0^1 (1 + r^{-2}) \log[F(r)F(-r)]\, \mathrm{d}r, \tag{4.20}$$

$$\liminf_n b_n \ge \frac{1}{2} \int_0^\pi \operatorname{Re} f(\theta)\, \mathrm{d}\theta. \tag{4.21}$$

The scalar integral includes its endpoints: the radial lower bound for R_n from the sampled reciprocal estimate and the upper bound from $\sqrt{P}R_n$ give a common logarithmic majorant at $r = 1$, while the even logarithm is $O(r^2)$ at zero. The Christoffel limit uses the compact-interior expansion at fixed bulk distance; the uniform lower bound in (3.1) controls the discarded endpoint atoms before the bulk distance tends to zero.

Define

$$\mathcal{B}(F) = \mathcal{A}(F) + \frac{1}{2} \int_0^\pi \operatorname{Re} \frac{F^*(e^{i\theta})}{\overline{F^*(e^{i\theta})}}\, \mathrm{d}\theta. \tag{4.22}$$

All these terms are finite. Combining (4.19)–(4.21) with the phase identity gives

$$\limsup_n c_n \le \mathcal{B}(F) - \liminf_n \Delta_n. \tag{4.23}$$

The original residual Δ_n cannot be discarded when seeking the sharp coefficient. The next section bounds it from below by retaining every fixed collection of the highest Legendre coefficients.

5 Residual energy and the sharp comparison

5.1 The residual in the highest Legendre modes

We retain a fixed number of the highest Legendre coefficients of the cardinal residuals before passing to a limit. This gives a nonnegative functional of the analytic limit and, in particular, proves its finiteness for the node sequences under consideration.

Throughout this section, $n \ge 2$ tends to infinity along a sequence of distinct nodes $x_{i,n} \in [-1, 1]$ for which $I_n \le 2$. We use the normalizations

$$q_n(x) = 2^{n-1} \prod_{i=1}^n (x - x_{i,n}), \qquad R_n(z) = 2z^n q_n\left(\frac{z + z^{-1}}{2} \right), \qquad P(z) = 1 - z^2.$$

Write

$$d_n = I_n - \sum_i \lambda_n(x_{i,n}) = \sum_i \|\ell_i - \lambda_n(x_{i,n})K_n(\cdot, x_{i,n})\|^2_{L^2(dx)}, \qquad \Delta_n = nd_n.$$

The finite estimates proved above give

$$0 \le \Delta_n \le D, \qquad \sup_n \|q_n\|_{L^2(dx)} < \infty, \qquad \sum_i \frac{1}{q_n'(x_{i,n})^2} \le \frac{J}{n}. \tag{5.1}$$

Here $D = 49 + 8\pi$ and $J = (2 + \sqrt{D})^2/2$. On the selected subsequence let $R_n \to F$ locally uniformly in the disk, where F is zero-free, has real Taylor coefficients, and $F(0) = 1$. Set

$$D_0 = P^{1/2}, \qquad A = D_0 F = \sum_{j\ge 0} a_j z^j, \qquad G = \frac{P}{F}, \qquad H_k(x) = \sum_{j=0}^{k} a_j U_{k-j}(x). \tag{5.2}$$

Every power of P is its analytic branch equal to one at zero, and U_j denotes the Chebyshev polynomial of the second kind. The identity

$$\int_0^{2\pi} |D_0(e^{i\theta})R_n(e^{i\theta})|^2 \frac{d\theta}{2\pi} = \frac{8}{\pi}\|q_n\|^2_{L^2(dx)}$$

and finite coefficient sums imply $A \in H^2$.

We use on this same subsequence the angular and reciprocal limits established above. With $\theta_i = \arccos x_{i,n}$ and $\nu_n = n^{-1}\sum_i \delta_{\theta_i}$, these are

$$\begin{aligned} &\nu_n \rightharpoonup \frac{d\theta}{\pi}, \qquad e^{2in\theta}\nu_n \rightharpoonup f(\theta)\frac{d\theta}{\pi}, \qquad |f| = 1, \qquad f = -\frac{F^*}{\overline{F^*}}, \\ &g_{i,n} = \frac{2ne^{i(1-n)\theta_i}}{q_n'(x_{i,n})}, \qquad \frac{1}{n}\sum_i |g_{i,n}|^2 \le 4J, \qquad g_n\nu_n \rightharpoonup G^*(e^{i\theta})\frac{d\theta}{\pi}, \qquad G \in H^2. \end{aligned} \tag{5.3}$$

The limits in this display are limits against continuous tests. The formula for $g_{i,n}$ includes endpoint nodes. A star denotes the almost-everywhere Hardy trace. No value of an arbitrary boundary representative is assigned to a moving node.

Lemma 5.1 (The finite top coefficients)**.** *Let $\psi_m = \sqrt{m+1/2}\,L_m$ be the ordinary $L^2(-1,1;dx)$-orthonormal Legendre polynomial. For every fixed $k \ge 0$, and all sufficiently large n,*

$$\langle \ell_i, \psi_{n-1-k}\rangle_{L^2(dx)} = \frac{\sqrt{\pi}}{q_n'(x_{i,n})}\big[H_k(x_{i,n}) + \varepsilon_{n,k}(x_{i,n})\big], \tag{5.4}$$

where $\varepsilon_{n,k}$ is a polynomial of degree at most k whose coefficients tend to zero. In particular, $\|\varepsilon_{n,k}\|_{L^\infty[-1,1]} \to 0$.

Proof. Put $b_m = 4^{-m}\binom{2m}{m}$. The Legendre generating function, or its three-term recurrence, gives

$$\sum_{m\ge 0} L_m(x)t^m = (1 - 2xt + t^2)^{-1/2}.$$

At $x = (z + z^{-1})/2$ its factorization yields

$$L_m\left(\frac{z + z^{-1}}{2}\right) = \sum_{j=0}^{m} b_j b_{m-j} z^{2j-m}. \tag{5.5}$$

The leading coefficient of L_m is $2^m b_m$. Therefore $Q_m^L = L_m/(2b_m)$ has leading Chebyshev coefficient one, and, for every fixed l with $2l < m$, its coefficient at T_{m-2l} is exactly

$$\frac{b_l b_{m-l}}{b_m} \longrightarrow b_l.$$

The odd defects vanish. Inverting this finite triangular connection shows that, for every fixed j, the coefficient of Q_{M-j}^L in T_M tends to d_j, where

$$\sum_{j\geq 0} d_j z^j = (1-z^2)^{1/2} = D_0(z).$$

Indeed the limiting convolution is the inverse of $\sum_{l\geq 0} b_l z^{2l} = P^{-1/2}$, and its diagonal is one. For each fixed defect this is a finite recursion.

The normalization is

$$Q_m^L = \frac{\psi_m}{2\sqrt{m+1/2}\, b_m}, \qquad \frac{1}{2\sqrt{m+1/2}\, b_m} \longrightarrow \frac{\sqrt{\pi}}{2}.$$

Here $b_m = C_m = 4^{-m}\binom{2m}{m}$, so the already proved refined Wallis estimate (4.11) gives $\sqrt{m}\, b_m \to \pi^{-1/2}$. Consequently,

$$\langle T_{m+j}, \psi_m \rangle_{L^2(dx)} \longrightarrow \frac{\sqrt{\pi}}{2} d_j \qquad (j \text{ fixed}). \tag{5.6}$$

Write $r_{j,n} = [z^j]R_n$. Reciprocity gives the exact expansion

$$q_n(x) = \sum_{j=0}^{n-1} r_{j,n} T_{n-j}(x) + \frac{r_{n,n}}{2}.$$

For real t, let $D_{n,t}(x) = [q_n(x) - q_n(t)]/(x-t)$. Comparing high Chebyshev coefficients in $(x-t)D_{n,t} = q_n(x) - q_n(t)$, using the three-term recurrence, gives its coefficient at T_{n-1-r}:

$$2\sum_{j=0}^{r} r_{j,n} U_{r-j}(t).$$

We may take $n \geq k+3$, so the constant-mode convention does not enter any coefficient being used. Only $r \leq k$ contributes to the projection onto ψ_{n-1-k}. Apply (5.6), use $r_{j,n} \to [z^j]F$, and convolve the finitely many coefficients. The resulting limit is $\sqrt{\pi}$ times the coefficient of z^k in

$$\frac{D_0(z)F(z)}{1-2tz+z^2} = \frac{A(z)}{1-2tz+z^2},$$

namely $\sqrt{\pi}H_k(t)$. This is coefficientwise convergence of polynomials in t of degree at most k, hence uniform on $[-1,1]$. Finally $D_{n,x_i} = q_n'(x_i)\ell_i$, which proves (5.4). □

Proposition 5.2 (The original residual penalty)**.** *For $\zeta = e^{i\theta}$, $0 < \theta < \pi$, let*

$$L_\zeta(w) = \frac{A(\zeta)(1-\zeta w)}{P(\zeta)} + \frac{\overline{A(\zeta)}(1-\bar{\zeta}w)}{\overline{P(\zeta)}} = \frac{A(\zeta)(1-\zeta w) - \zeta^2\overline{A(\zeta)}(1-w/\zeta)}{P(\zeta)}, \tag{5.7}$$

and define

$$Q_\zeta(w) = \frac{A(w) - L_\zeta(w)}{(1-\zeta w)(1-\bar{\zeta}w)}, \qquad \mathcal{G}(F) = \frac{1}{4}\int_0^\pi |G(e^{i\theta})|^2 \|Q_{e^{i\theta}}\|_{H^2}^2\, d\theta. \tag{5.8}$$

The norm in this definition is the supremum of radial squared norms and is allowed to be infinite for arbitrary factors. Under (5.1)–(5.3),

$$\liminf_{n\to\infty} \Delta_n \geq \mathcal{G}(F), \qquad \mathcal{G}(F) \leq D < \infty. \tag{5.9}$$

Proof. For a fixed k, put $m = n-1-k$. On every fixed compact angular subinterval of $(0,\pi)$, the Legendre specialization of the Darboux formula [6, Eq. 18.15.4_5] is

$$\psi_m(\cos\theta) = \sqrt{\frac{2}{\pi\sin\theta}}\cos((n-\tfrac{1}{2}-k)\theta - \tfrac{\pi}{4}) + o(1),$$

uniformly there. The exact diagonal identity

$$K_n(\cos\theta,\cos\theta) = \frac{n^2}{2\sin^2\theta}[L_{n-1}^2 - 2\cos\theta L_n L_{n-1} + L_n^2]$$

and the same two undifferentiated Darboux expansions show that $n\lambda_n(\cos\theta) \to \pi\sin\theta$ uniformly on this interval. The leading trigonometric quadratic is $\sin^2\theta$, independently of the oscillatory phase.

Set

$$c_{i,k} = \langle \ell_i, \psi_m\rangle - \lambda_n(x_i)\psi_m(x_i), \qquad h_{i,n,k} = ne^{i(1-n)\theta_i}c_{i,k}.$$

The error in Lemma 5.1 vanishes in the actual discrete L^2 norm, since

$$\frac{1}{n}\sum_i \left|\frac{n\varepsilon_{n,k}(x_i)}{q_n'(x_i)}\right|^2 \le J\|\varepsilon_{n,k}\|_\infty^2 \longrightarrow 0.$$

The reciprocal limit therefore gives the weak limit of the first term of $h_{i,n,k}$ as $\sqrt{\pi}GH_k/2$. Splitting the cosine of the second term into two exponentials, and using the conjugate phase measures in (5.3), gives its limit as

$$\sqrt{\frac{\pi\sin\theta}{2}}\left[e^{i(1/2-k)\theta - i\pi/4} + e^{i(k+3/2)\theta + i\pi/4}\overline{f(\theta)}\right].$$

Thus the weak limit on every fixed bulk interval is

$$E_k(\theta) = \frac{\sqrt{\pi}}{2}G(e^{i\theta})H_k(\cos\theta) - \sqrt{\frac{\pi\sin\theta}{2}}\left[e^{i(1/2-k)\theta - i\pi/4} + e^{i(k+3/2)\theta + i\pi/4}\overline{f(\theta)}\right]. \tag{5.10}$$

For every fixed K, Bessel's inequality for the distinct modes $\psi_{n-1},\dots,\psi_{n-1-K}$ gives

$$\Delta_n \ge \sum_{k=0}^{K}\frac{1}{n}\sum_i |h_{i,n,k}|^2.$$

The lower bound for the moving measures follows directly from duality. For a continuous vector test φ supported in a compact bulk interval,

$$\int |h_n|^2\,d\nu_n \ge 2\,\mathrm{Re}\int h_n\overline{\varphi}\,d\nu_n - \int |\varphi|^2\,d\nu_n.$$

Passing to the weak limits and taking the supremum over these tests proves the lower bound by their density in the corresponding L^2 space. Increase the bulk intervals to $(0,\pi)$; the discarded endpoint terms were nonnegative. We obtain

$$\liminf_n \Delta_n \ge \frac{1}{\pi}\sum_{k=0}^{K}\int_0^\pi |E_k(\theta)|^2\,d\theta. \tag{5.11}$$

Here K remains fixed until after the limit in n.

The affine function in (5.7) satisfies $L_\zeta(\zeta) = A(\zeta)$ and $L_\zeta(\bar\zeta) = \overline{A(\zeta)}$. Summing (5.10) geometrically gives the exact identity

$$\sum_{k\ge 0}E_k(\theta)w^k = \frac{\sqrt{\pi}}{2}G(\zeta)Q_\zeta(w), \qquad |w|<1. \tag{5.12}$$

For the constants and phases, one uses

$$\sum_{k\ge 0} H_k(x)w^k = \frac{A(w)}{1-2xw+w^2}, \qquad D_0(\zeta) = \sqrt{2\sin\theta}\, e^{i\theta/2-i\pi/4}, \qquad \overline{D_0(\zeta)} = i\zeta^{-1}D_0(\zeta).$$

The two geometric numerators in (5.10) are $D_0(\zeta)$ and $i\zeta D_0(\zeta)\overline{f}$. The identities $f = -F/\bar{F}$ and $G = P/F$ turn them respectively into $GA(\zeta)/P(\zeta)$ and $G\overline{A(\zeta)}/\overline{P(\zeta)}$, proving (5.12).

Radial Parseval, followed by monotone summation of the nonnegative coefficients, identifies

$$\frac{1}{\pi}\sum_{k\ge 0}\int_0^\pi |E_k|^2\, d\theta = \mathcal{G}(F).$$

Letting $K\to\infty$ in (5.11) proves (5.9). Since $G\neq 0$ almost everywhere, finite $\mathcal{G}(F)$ also gives $Q_\zeta \in H^2$ for almost every θ. Its trace agrees with the boundary quotient away from the two interpolation points. Consequently the same functional has the nonnegative integral form

$$\mathcal{G}(F) = \frac{1}{4}\int_0^\pi |G(e^{i\theta})|^2 \int_0^{2\pi} \frac{|A(e^{i\vartheta}) - L_{e^{i\theta}}(e^{i\vartheta})|^2}{|e^{i\vartheta}-e^{i\theta}|^2|e^{i\vartheta}-e^{-i\theta}|^2}\frac{d\vartheta}{2\pi}\, d\theta. \tag{5.13}$$

The Hardy membership is obtained from the mode bound, not merely from cancellation of the two boundary zeros. □

5.1.1 Normalization examples

For the Legendre Gaussian factor $F = P^{-1/2}$, one has $A = 1$, $G = P^{3/2}$, $L_\zeta = 1$, and $Q_\zeta = 0$. Thus every E_k vanishes and $\mathcal{G}(P^{-1/2}) = 0$. More generally an affine $A = 1 + bz$, whenever its companion $G = P^{3/2}/A$ is admissible, has zero penalty. Conversely, zero penalty and nonzero G force A to be affine, since one of the almost-everywhere vanishing analytic quotients already identifies A with an affine function.

For $F = 1$, write

$$D_0(z) = \sum_{j\ge 0} d_j z^j, \qquad d_0 = 1, \quad d_{2m} = -\frac{b_m}{2m-1}\ (m\ge 1), \quad d_{2m+1} = 0.$$

For a polynomial A, polynomial division gives

$$[w^k]Q_\zeta(w) = \sum_{j\ge k+2} a_j U_{j-k-2}(\cos\theta).$$

For $A = D_0$, the same formula follows by its absolutely summable coefficients, or by the weighted orthogonality limit. Since $|G|^2 = 4\sin^2\theta$, sine orthogonality gives

$$\mathcal{G}(1) = \frac{\pi}{2}\sum_{m\ge 1}(2m-1)d_{2m}^2 = \frac{\pi}{2}\sum_{m\ge 1}\frac{b_m^2}{2m-1}.$$

The recurrence $b_m/b_{m-1} = (2m-1)/(2m)$ implies

$$\frac{b_m^2}{2m-1} = (2m-1)b_{m-1}^2 - (2m+1)b_m^2.$$

This telescopes, and $(2m+1)b_m^2 \to 2/\pi$; hence $\mathcal{G}(1) = \pi/2 - 1$. Finally, for $F = P^{1/2}$, $A = P$, $G = P^{1/2}$, and $Q_\zeta = -1$, so that $\mathcal{G}(P^{1/2}) = \frac{1}{4}\int_0^\pi |P(e^{i\theta})|\, d\theta = 1$.

5.2 Fourier–Bessel evaluation of the reference energy

The constant needed below can be evaluated without any node construction. We distinguish Euler's gamma function $\Gamma_{\mathrm{E}}(s) = \int_0^\infty t^{s-1}e^{-t}\,dt$, $s > 0$, from the penalty $\mathcal{G}$, and write

$$c_* = \frac{\Gamma_{\mathrm{E}}(1/4)^2}{8\Gamma_{\mathrm{E}}(3/4)^2}. \tag{5.14}$$

For $|x| < 1$, set $f_0(x) = (1-x^2)^{1/4}$, and extend it by zero to $\mathbb{R}$.

Proposition 5.3. *The zero extension f_0 has finite energy, and*

$$\frac{1}{4\pi}\iint_{\mathbb{R}^2} \frac{[f_0(x) - f_0(t)]^2}{(x-t)^2}\,dx\,dt = c_*. \tag{5.15}$$

The part of this energy for pairs with one point outside $[-1,1]$ equals one.

Proof. All squared-difference integrals are nonnegative extended integrals until their finiteness is established. The value on the diagonal is immaterial. Near an endpoint, the power model with $\beta = 1/4$ satisfies

$$\int_0^a \int_0^a \frac{(s^\beta - t^\beta)^2}{(s-t)^2}\,dt\,ds = 2\left(\int_0^a s^{2\beta-1}\,ds\right)\left(\int_0^1 \frac{(1-u^\beta)^2}{(1-u)^2}\,du\right) < \infty.$$

The smooth positive factor in f_0 preserves this bound: split the product difference into its power difference and its bounded Lipschitz-factor difference, then apply $(u+v)^2 \le 2u^2 + 2v^2$. Away from the endpoint corners, local difference quotients are bounded, while separated-corner terms have an integrable denominator bound. Exterior pairs give exactly

$$\frac{1}{2\pi}\int_{-1}^1 \sqrt{1-x^2}\left(\frac{1}{1-x} + \frac{1}{1+x}\right)\,dx = 1. \tag{5.16}$$

This proves finite energy without any information about Jacobi nodes.

For clarity, the Fourier normalization and its energy conversion are given explicitly. Let

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-i\xi x}\,dx, \qquad \|f\|_2^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\xi)|^2\,d\xi.$$

The second identity is real-line Plancherel after the frequency change $\xi = 2\pi\eta$ from the unitary convention [10, Theorem 43 and Exercise 44]. In this application f_0, and every translation difference of f_0, belong to $L^1 \cap L^2$, so the Fourier transforms agree with their integral definitions.

The elementary Dirichlet integral fixes the remaining constant. For $a > 0$, define

$$J(a) = \int_0^\infty e^{-at}\frac{\sin t}{t}\,dt.$$

Differentiation under the exponentially dominated integral gives $J'(a) = -(1+a^2)^{-1}$; the bound $|J(a)| \le a^{-1}$ at infinity then gives $J(a) = \arctan(1/a)$. The bounded primitive of $\sin t$, with the decreasing factor e^{-at}/t, bounds the tail on $[R,\infty)$ by C/R, uniformly for $a \ge 0$. It follows that $J(0) = \pi/2$. Integration by parts, with vanishing boundary terms, gives

$$\int_0^\infty \frac{1-\cos t}{t^2}\,dt = \frac{\pi}{2}, \qquad \int_{\mathbb{R}} \frac{2-2\cos(\xi s)}{s^2}\,ds = 2\pi|\xi|.$$

Apply Plancherel to $f_0(\cdot + s) - f_0$, whose Fourier multiplier is $e^{i\xi s} - 1$. Tonelli's theorem for the nonnegative s^{-2} integral [8, Theorem 4] yields the extended identity

$$\iint_{\mathbb{R}^2} \frac{[f_0(x) - f_0(t)]^2}{(x-t)^2}\,dx\,dt = \int_{\mathbb{R}} |\xi|\,|\widehat{f_0}(\xi)|^2\,d\xi. \tag{5.17}$$

Poisson's Bessel representation [6, Eq. 10.9.4], at $\nu = 3/4 > -1/2$, gives for $\xi \neq 0$

$$\widehat{f}_0(\xi) = \sqrt{\pi}\,\Gamma_{\mathrm{E}}(5/4)\left(\frac{2}{|\xi|}\right)^{3/4} J_{3/4}(|\xi|). \tag{5.18}$$

Its value at zero is removable and does not affect the energy. The Bessel product integral [6, Eq. 10.22.57], with $a = 1$, $\mu = \nu = 3/4$, and $\lambda = 1/2$, is applicable because $0 < 1/2 < \mu + \nu + 1 = 5/2$. It gives

$$\int_0^\infty J_{3/4}(t)^2 t^{-1/2}\,dt = \frac{\sqrt{2}}{\Gamma_{\mathrm{E}}(3/4)^2}. \tag{5.19}$$

Substitution in (5.17) therefore gives

$$\frac{1}{4\pi}\int_{\mathbb{R}} |\xi|\,|\widehat{f}_0(\xi)|^2\,d\xi = \frac{2\Gamma_{\mathrm{E}}(5/4)^2}{\Gamma_{\mathrm{E}}(3/4)^2} = \frac{\Gamma_{\mathrm{E}}(1/4)^2}{8\Gamma_{\mathrm{E}}(3/4)^2}.$$

The last equality uses $\Gamma_{\mathrm{E}}(5/4) = \Gamma_{\mathrm{E}}(1/4)/4$, obtained by integration by parts in the positive-real gamma integral. This proves (5.15) independently of the node construction below. □

5.3 A direct entropy identity for finite penalty

This section applies directly to the possibly rough analytic factors obtained from the nodes. The finiteness of the residual penalty supplies the boundary difference quotient needed for a positive entropy identity. A common positive cutoff then compares that identity with the reference energy of Section 5.2.

Let F be zero-free and analytic in the disk, with real Taylor coefficients and $F(0) = 1$, and suppose

$$A = P^{1/2}F \in H^2, \qquad G = P/F \in H^2, \qquad AG = P^{3/2}, \qquad \mathcal{G}(F) < \infty, \quad P(z) = 1 - z^2. \tag{5.20}$$

The penalty and the interpolating quotient are those of (5.7)–(5.8). Write, for $\zeta = e^{i\theta}$, $0 < \theta < \pi$,

$$\begin{gathered} x = \cos\theta, \qquad y_x = \sqrt{1-x^2}, \qquad \rho(x) = |A(\zeta)|^2, \qquad \phi(\theta) = \operatorname{Im}\log A(\zeta), \\ M(x) = |F(\zeta)|^2, \qquad f_0(x) = \sqrt{y_x}, \qquad h(x) = \frac{M(x)}{f_0(x)}. \end{gathered} \tag{5.21}$$

Boundary values are taken almost everywhere. The logarithm is the analytic branch with $\log A(0) = 0$. In particular ϕ is the value of its boundary imaginary part, rather than an angular derivative or a principal argument reduced modulo 2π. The scalar functions occurring in the coefficient reduction are

$$\begin{aligned} \mathcal{A}(F) &= -\frac{1}{2}\int_0^1 (1 + r^{-2})\log[F(r)F(-r)]\,dr, \\ \mathcal{B}(F) &= \mathcal{A}(F) + \frac{1}{2}\int_0^\pi \operatorname{Re}\frac{F(e^{i\theta})}{\overline{F(e^{i\theta})}}\,d\theta. \end{aligned} \tag{5.22}$$

The radial integral will be shown to converge under (5.20) itself.

Theorem 5.4 (The finite-penalty entropy identity)**.** *Under* (5.20)*,*

$$\mathcal{B}(F) - \mathcal{G}(F) = c_* - \mathcal{L}[h], \tag{5.23}$$

where

$$\mathcal{L}[h] = \frac{1}{4\pi}\iint_{(-1,1)^2} \sqrt{y_x y_t}\,\frac{h(t)/h(x) + h(x)/h(t) - 2}{(x-t)^2}\,dx\,dt \quad \textit{satisfies} \quad 0 \le \mathcal{L}[h] < \infty. \tag{5.24}$$

Thus $\mathcal{B}(F) - \mathcal{G}(F) \le c_*$*. Equality holds if and only if* $F = P^{1/4}$*.*

We first establish the logarithmic and two-point identities in the full factor class. The measure-theoretic passages below use monotone convergence, Fatou's lemma, dominated convergence, and Tonelli or absolutely integrable Fubini [8, Theorems 3–4]. The interval, circle, and arcsine measures are their standard complete Lebesgue versions. When the completed product is used, both factor spaces are complete and sigma-finite, as required for that version of Fubini's theorem. All null zero sets and diagonals may be assigned arbitrary values without changing any integral.

5.3.1 Logarithms and Poisson defects

Lemma 5.5. *The factors A and G have nonzero traces almost everywhere, $\log|A|^2$ and $\log|G|^2$ belong to L^2 of the circle, and their logarithmic Poisson defects vanish. Their normalized analytic logarithms, as well as $\log F$, belong to H^2.*

Proof. For H^2 we use normalized circle measure $d\theta/(2\pi)$. Radial convergence in L^2 and almost-everywhere boundary convergence are the usual Hardy trace results [7, Theorem 5.8 and Corollary 5.26]. Apply the L^2 trace convergence to A and G. Cauchy–Schwarz gives convergence of their product in L^1, so $A^*G^* = (P^{3/2})^*$. The latter is nonzero except at the two endpoints. Set $\sigma = |G^*|^2$, and extend ρ to the full circle by conjugate reflection. Then

$$\log\rho + \log\sigma = 3\log|P|.$$

Both ρ and σ are integrable. The inequality $(\log^+ s)^2 \le Cs$ gives square-integrability of their positive logarithms. Moreover,

$$\log^-\rho \le \log^+\sigma + 3|\log|P||, \qquad \log^-\sigma \le \log^+\rho + 3|\log|P||.$$

The function $\log|1 - e^{2i\theta}|$ is in circle L^2; hence both full real logarithms are in L^2.

It is necessary to rule out a Poisson defect, rather than infer outerness from zero-freeness. Fix $|z| < r < 1$, and let $\mathsf{P}_z(e^{it}) = (1-|z|^2)/|e^{it}-z|^2$ denote the Poisson kernel. Since A is zero-free, its logarithmic modulus is harmonic, and reproduction on the inner circle gives

$$\log|A(z)| = \int_0^{2\pi} \mathsf{P}_{z/r}(e^{it})\log|A(re^{it})|\frac{dt}{2\pi}.$$

The function $u \mapsto \log^+|u|$ is Lipschitz with constant one. Radial L^2 convergence therefore gives L^1 convergence of its values. The Poisson kernels in this display converge uniformly as $r \uparrow 1$; Fatou's lemma applied to the negative parts then proves

$$\log|A(z)| \le \int_0^{2\pi} \mathsf{P}_z(e^{it})\log|A^*(e^{it})|\frac{dt}{2\pi}.$$

The same argument applies to G. These two nonnegative defects sum to zero: their sum is the defect for $P^{3/2}$, and

$$\log P^{3/2}(z) = -\frac{3}{2}\sum_{k\ge1}\frac{z^{2k}}{k} \in H^2$$

has exactly its boundary real part as Poisson data. Thus each individual defect vanishes at every point of the disk. This is the precise outer property used here.

Finally, compare Fourier coefficients of the Poisson representation of $\operatorname{Re}\log A$ with its L^2 real boundary part $\frac{1}{2}\log\rho$. The normalized analytic logarithm has square-summable Taylor coefficients, so it belongs to H^2. This coefficient argument does not require a prior trace for the logarithm. Its H^2 trace then supplies that trace, and in particular $\phi \in L^2(0,\pi)$. The same argument applies to G, and $\log F = \log A - \frac{1}{2}\log P \in H^2$. □

5.3.2 A two-point identity without finite-part splitting

Lemma 5.6. *Let $\Phi(s) = s - 1 - \log s$, $s > 0$, and put $d\nu(t) = dt/(\pi y_t)$. Write the interpolant as*

$$L_\zeta(w) = a(x) + wb(x), \qquad b(x) = \frac{\operatorname{Im} A(\zeta)}{y_x}, \qquad a(x) = \operatorname{Re} A(\zeta) - xb(x),$$

so that a, b are real and $\rho = a^2 + 2xab + b^2$. For almost every $x = \cos\theta$,

$$\begin{aligned} S(x) := \|Q_\zeta\|_{H^2}^2 = \frac{\rho(x)}{4}\int_{-1}^{1} \frac{\Phi(\rho(t)/\rho(x))}{(x-t)^2}\, d\nu(t) \\ + \frac{\rho(x)x\phi(\theta)}{2y_x^3} - \frac{xab + b^2}{2y_x^2}. \end{aligned} \tag{5.25}$$

Proof. Finite $\mathcal{G}(F)$ and nonvanishing of G show that $Q_\zeta \in H^2$ for almost every θ. Fix such a point with all the traces in Lemma 5.5. For brevity, write $y = y_x$, $z_r = r\zeta$, $s_r = 1 - r^2$, and

$$K_r(w) = \frac{1}{|w - r\zeta|^2 |w - r\bar{\zeta}|^2}, \qquad \alpha_r = \frac{\zeta}{2iy(1-r^2)(1-r^2\zeta^2)}.$$

On the unit circle,

$$|w - r\zeta|^2 = r|w - \zeta|^2 + (1-r)^2, \qquad K_r \le r^{-2} K_1.$$

The boundary quotient defining Q_ζ consequently gives

$$N_r := \int_0^{2\pi} |A(e^{it}) - L_\zeta(e^{it})|^2 K_r(e^{it}) \frac{dt}{2\pi} \longrightarrow S(x). \tag{5.26}$$

We record the rational integration identities, including the term with a negative power. For any real-coefficient $H \in H^2$,

$$\begin{aligned} \int_0^{2\pi} H(w) K_r(w) \frac{dt}{2\pi} &= 2\operatorname{Re}[\alpha_r H(z_r)], \\ \int_0^{2\pi} A(w) w^{-1} K_r(w) \frac{dt}{2\pi} &= 2\operatorname{Re}[\alpha_r A(z_r)/z_r], \\ \int_0^{2\pi} \log \rho(w) K_r(w) \frac{dt}{2\pi} &= 4\operatorname{Re}[\alpha_r \log A(z_r)], \qquad w = e^{it}. \end{aligned} \tag{5.27}$$

To see these formulas directly, the rational function $K_r(w)/w$ has its two inside poles at $r\zeta, r\bar{\zeta}$; the coefficients of its simple partial fractions there are $\alpha_r, \overline{\alpha_r}$. Its other two poles are outside the unit circle. For a polynomial H, the geometric series for $(w-a)^{-1}$ integrates $H(w)/(w-a)$ to $H(a)$ if $|a| < 1$, and to zero if $|a| > 1$. Polynomial approximation in H^2 extends this identity because the fixed circle kernels are bounded and point evaluation is bounded. Replacing H by A/w changes the rational factor to $K_r(w)/w^2$. It has no pole at zero, since K_r has a w^2 numerator, and divides the two inside coefficients by $r\zeta$ and $r\bar{\zeta}$, respectively. This proves the second identity without an extra zero contribution. The last identity follows from $\log\rho = 2\operatorname{Re}\log A$ and Lemma 5.5.

The required local rate follows from the quotient, not from a smoothness assumption. If $V \in H^2$, its Taylor coefficients give $|V(z)| \le \|V\|_{H^2}/\sqrt{1-|z|^2}$. Splitting V into a polynomial and an arbitrarily small H^2 tail therefore proves

$$\sqrt{1-r^2}\,|V(r\zeta)| \longrightarrow 0 \quad \text{at each fixed } \zeta \in \mathbb{T}.$$

Here

$$\frac{A(w) - A(\zeta)}{w - \zeta} = b + (w - \bar{\zeta}) Q_\zeta(w),$$

so $|A(z_r) - A(\zeta)|^2/(1-r) \to 0$. Set $c = A(\zeta) \neq 0$ and $\ell = \log A(\zeta)$, using the normalized logarithmic trace. Taylor expansion of the exponential, together with this rate and the radial trace of $\log A$, gives

$$A(z_r) = c[1 + \log A(z_r) - \ell] + o(1-r). \tag{5.28}$$

Let

$$M_r = \int_0^{2\pi} \rho(x)\Phi\left(\frac{\rho(w)}{\rho(x)}\right) K_r(w)\frac{dt}{2\pi}.$$

For $r < 1$ all the integrals just used are finite. Expanding $|A - a - bw|^2$ and using (5.27) gives the exact expression

$$\begin{aligned} N_r - M_r = {} & 4\rho \operatorname{Re}\alpha_r + 4ab \operatorname{Re}[\alpha_r(z_r - x)] \\ & - 4\operatorname{Re}[\alpha_r(a + b/z_r)A(z_r)] + 4\rho \operatorname{Re}[\alpha_r \log A(z_r)] - 2\rho \log\rho(x) \operatorname{Re}\alpha_r. \end{aligned}$$

The elementary expansions required to take its limit are

$$\begin{gathered} \alpha_r = \frac{1}{4y^2 s_r} + \frac{1}{8y^2} - \frac{ix}{8y^3} + O(s_r), \\ \operatorname{Re}[\alpha_r(z_r - x)] \longrightarrow 0, \qquad \operatorname{Re}\left[\alpha_r cb\left(\frac{1}{z_r} - \frac{1}{\zeta}\right)\right] \longrightarrow \frac{b(ax+b)}{8y^2}. \end{gathered} \tag{5.29}$$

Inserting (5.28) and $a + b/\zeta = \bar{c}$, the constant and logarithmic real parts cancel. The surviving logarithmic term is $-4\rho\phi \operatorname{Im}\alpha_r$. Equations (5.29) therefore give

$$N_r - M_r \longrightarrow \frac{\rho x\phi}{2y^3} - \frac{xab + b^2}{2y^2}. \tag{5.30}$$

All remainders tend to zero, since $\alpha_r = O((1-r)^{-1})$ and the error in (5.28) is $o(1-r)$.

The integrand defining M_r is nonnegative. Fatou's lemma, (5.26), and (5.30) first show that its K_1 integral is finite. The inequality $K_r \le r^{-2}K_1$ then gives dominated convergence for M_r. Finally, on the circle $K_1(w) = 1/[4(x-t)^2]$, where $t = \operatorname{Re} w$, and conjugate reflection turns normalized circle integration into $d\nu(t)$. This proves (5.25) with no splitting of divergent finite parts. □

5.3.3 Integrated entropy and the scalar phase

Define the nonnegative entropy energy

$$\mathcal{T}(\rho) = \frac{1}{2\pi}\iint_{(-1,1)^2} \frac{y_x^2}{y_t}\,\frac{\Phi(\rho(t)/\rho(x))}{(x-t)^2}\,dx\,dt, \tag{5.31}$$

and the absolutely convergent quantities

$$Z = \int_0^\pi \cos\theta\,\phi(\theta)\,d\theta, \qquad C_0 = \int_{-1}^1 \frac{xab + b^2}{\rho(x)}\,dx.$$

For Z this follows from $\phi \in L^2$. For C_0, the useful cancellation is

$$\operatorname{Re}\frac{F}{\bar{F}} = \frac{y_x(a^2 - b^2)}{\rho}, \qquad \frac{xab + b^2}{\rho} = \frac{1}{2} - \frac{1}{2y_x}\operatorname{Re}\frac{F}{\bar{F}}. \tag{5.32}$$

In particular the last integrand is bounded in modulus by $1/2 + 1/(2y_x)$. Since $|G|^2 = 8y_x^3/\rho$, the measurable substitution $x = \cos\theta$ in (5.8) gives

$$\mathcal{G}(F) = 2\int_{-1}^1 \frac{y_x^2}{\rho(x)} S(x)\,dx.$$

Integrate the nonnegative term in (5.25) first and then its absolutely integrable corrections. It follows that

$$\mathcal{G}(F) = \mathcal{T}(\rho) + Z - C_0, \qquad \mathcal{T}(\rho) < \infty. \tag{5.33}$$

To identify the entire scalar integral in (5.22), write $\log F(z) = \sum_{j\ge1} c_j z^j$. These coefficients are real and square summable by Lemma 5.5. Their even part vanishes to order two at zero. Near $r = 1$, the H^2 point-evaluation bound is $O((1-r)^{-1/2})$, which is integrable. More explicitly, termwise integration is absolutely justified by

$$\sum_{m\ge1} |c_{2m}| \left(\frac{1}{2m-1} + \frac{1}{2m+1} \right) < \infty,$$

using Cauchy–Schwarz. Thus

$$\mathcal{A}(F) = -\sum_{m\ge1} c_{2m} \left(\frac{1}{2m-1} + \frac{1}{2m+1} \right) = -\int_0^\pi \cos\theta \ \operatorname{Im} \log F(e^{i\theta})\, d\theta. \tag{5.34}$$

The last equality follows first for polynomials by integrating $\cos\theta \sin(j\theta)$, and then by L^2 convergence of the Hardy trace. On the upper circle, $\operatorname{Im} \log P(e^{i\theta}) = \theta - \pi/2$, hence $\operatorname{Im} \log F = \phi - \theta/2 + \pi/4$. Since $\int_0^\pi \theta \cos\theta \, d\theta = -2$,

$$\mathcal{A}(F) = -Z - 1, \qquad \frac{1}{2}\int_0^\pi \operatorname{Re}(F/\bar{F})\, d\theta = 1 - C_0.$$

Combining these identities with (5.33) gives

$$\mathcal{B}(F) = -Z - C_0, \qquad \mathcal{B}(F) - \mathcal{G}(F) = -\mathcal{T}(\rho) - 2Z. \tag{5.35}$$

5.3.4 The common cutoff and its logarithmic kernel

For $\varepsilon > 0$, put

$$K_\varepsilon(x,t) = \frac{1}{(x-t)^2 + \varepsilon^2},$$

and let $\mathcal{T}_\varepsilon$ and $\mathcal{L}_\varepsilon$ be obtained from (5.31) and (5.24) by replacing the inverse-square factor with K_ε. Both are nonnegative and increase to their uncut versions as $\varepsilon \downarrow 0$. Every term in the finite-cutoff algebra below is absolutely integrable. For the ratio terms this follows, apart from ε^{-2}, from

$$\int_{-1}^1 \frac{y_x^2}{\rho(x)}\, dx = \frac{1}{8}\int_0^\pi |G(e^{i\theta})|^2\, d\theta < \infty, \qquad \int_{-1}^1 \frac{\rho(t)}{y_t}\, dt = \int_0^\pi |A(e^{i\theta})|^2\, d\theta < \infty. \tag{5.36}$$

The logarithmic terms are integrable because $\log\rho \in L^2(d\nu)$.

Lemma 5.7 (The limiting logarithmic functional)**.** *For real $u \in L^2(d\nu)$, set*

$$\operatorname{Lin}_\varepsilon(u) = -\frac{1}{2\pi} \iint_{(-1,1)^2} \frac{y_x^2}{y_t}[u(t) - u(x)] K_\varepsilon(x,t)\, dx\, dt.$$

Then

$$\operatorname{Lin}_\varepsilon(u) \longrightarrow \operatorname{Lin}_0(u) := \int_{-1}^1 u(t) k_0(t) \frac{dt}{y_t}, \qquad k_0(t) = \frac{1}{\pi}\left[2 - t \log \frac{1+t}{1-t} \right]. \tag{5.37}$$

If H is a real-coefficient analytic function with its normalized logarithm in H^2, and $u = \log|H^*|^2$, then

$$\operatorname{Lin}_0(u) = -2\int_0^\pi \cos\theta \ \operatorname{Im} \log H(e^{i\theta})\, d\theta. \tag{5.38}$$

Proof. Symmetrizing only the finite, absolutely integrable quantities gives

$$\mathrm{Lin}_\varepsilon(u) = \int_{-1}^{1} u(t)k_\varepsilon(t)\frac{dt}{y_t}, \qquad k_\varepsilon(t) = \frac{1}{2\pi}\int_{-1}^{1}\left(\frac{y_t^3}{y_x} - y_x^2\right)K_\varepsilon(x,t)\,dx.$$

The exact kernel is

$$\begin{aligned} 2\pi k_\varepsilon(t) = {} & \frac{\pi y_t^3}{\varepsilon}\,\mathrm{Re}[1-(t+i\varepsilon)^2]^{-1/2} \\ & - \frac{y_t^2+\varepsilon^2}{\varepsilon}\left[\arctan\frac{1-t}{\varepsilon} + \arctan\frac{1+t}{\varepsilon}\right] \\ & + t\log\frac{(1-t)^2+\varepsilon^2}{(1+t)^2+\varepsilon^2} + 2. \end{aligned} \tag{5.39}$$

The square root has positive real part. The polynomial contribution is obtained by expanding $1-x^2$ about $x=t$.

Here is a direct derivation of the other contribution with its branch fixed. For $w = t+i\varepsilon$, let $D = \sqrt{1-w^2}$ have positive real part, and put $a = w - iD$. Its real part satisfies

$$(\mathrm{Re}\,D)^2 = \frac{|1-w^2| + 1 - t^2 + \varepsilon^2}{2} > \varepsilon^2, \qquad |1-w^2|^2 - (|w|^2-1)^2 = 4\varepsilon^2.$$

Also $(w-iD)(w+iD) = 1$, so $a + a^{-1} = 2w$, and $\mathrm{Im}\,a < 0$. Writing $a = re^{i\beta}$, the identity $\varepsilon = (r - r^{-1})\sin\beta/2$ shows $0 < r < 1$. The two uniformly absolutely convergent geometric series now give

$$\frac{1}{\pi}\int_0^\pi \frac{d\theta}{1-2a\cos\theta+a^2} = \frac{1}{1-a^2}, \qquad \int_{-1}^{1}\frac{dx}{y_x(w-x)} = \frac{2\pi a}{1-a^2} = \frac{\pi}{iD}.$$

Taking imaginary parts yields

$$\int_{-1}^{1}\frac{dx}{y_x[(x-t)^2+\varepsilon^2]} = \frac{\pi}{\varepsilon}\,\mathrm{Re}[1-(t+i\varepsilon)^2]^{-1/2}.$$

All integrals here are absolutely convergent for fixed $\varepsilon > 0$, so this derivation requires no continuation of an endpoint integral.

For each fixed $-1 < t < 1$, (5.39) gives $k_\varepsilon(t) \to k_0(t)$. We need uniform endpoint control to pair this limit with rough u. Put $h_t = y_t^2$. If $0 < \varepsilon \le h_t/4$, expand the real part of

$$\left[1 + \frac{\varepsilon^2 - 2it\varepsilon}{h_t}\right]^{-1/2}.$$

Its difference from one is $O(\varepsilon^2/h_t^2)$. The first term of (5.39) differs from $\pi h_t/\varepsilon$ by $O(\varepsilon/h_t)$, hence by a uniform constant. For the second term, use

$$\arctan\frac{1-t}{\varepsilon} + \arctan\frac{1+t}{\varepsilon} = \pi - \arctan\frac{\varepsilon}{1-t} - \arctan\frac{\varepsilon}{1+t}.$$

After multiplication by h_t/ε, the two corrections are bounded by $h_t/(1-t) + h_t/(1+t) = 2$. The additional ε^2 contribution is at most $\pi\varepsilon$. Thus the two divergent leading terms cancel with a uniformly bounded remainder.

If $h_t/4 \le \varepsilon \le 1$, then $|[1-(t+i\varepsilon)^2]^{-1/2}| \le h_t^{-1/2}$. The first term is at most $\pi h_t/\varepsilon \le 4\pi$, and the second is at most $\pi(h_t/\varepsilon + \varepsilon) \le 5\pi$. Finally, adding ε^2 to numerator and denominator of the logarithmic ratio decreases the absolute logarithm. Both regions therefore give the common bound

$$|k_\varepsilon(t)| \le C[1 + |\log(1-t^2)|], \qquad 0 < \varepsilon \le 1. \tag{5.40}$$

The right side is in $L^2(d\nu)$. Dominated convergence gives $k_\varepsilon \to k_0$ in that space, and Cauchy–Schwarz proves (5.37).

For the phase formula, cosine polynomials suffice by L^2 continuity. One may compute their pairings with k_0 using the $L^2(0,\pi)$ Fourier expansion

$$\log \frac{1+\cos\theta}{1-\cos\theta} = 4 \sum_{\substack{j\ge 1\\ j \text{ odd}}} \frac{\cos(j\theta)}{j}.$$

This follows by taking real parts of the two analytic logarithm series for $1+z$ and $1-z$, whose coefficients are square summable. Cosine orthogonality gives

$$\int_0^\pi k_0(\cos\theta)\cos(j\theta)\,d\theta = \begin{cases} -[(j-1)^{-1}+(j+1)^{-1}], & j \ge 2 \text{ even},\\ 0, & j = 0 \text{ or } j \text{ odd}. \end{cases}$$

For $\log H(z) = c_0 + \sum_{j\ge1} c_j z^j$, these are exactly the pairings of $-2\cos\theta$ with $\sum_{j\ge1} c_j \sin(j\theta)$, since $u = 2c_0 + 2\sum_{j\ge1} c_j \cos(j\theta)$. Both functionals extend continuously to L^2: on the phase side this follows from Parseval and the bounded multiplier for circle harmonic conjugation [10, Theorem 21 and Corollary 22]. This proves (5.38) without exchanging rough principal values. □

The substitutions used here and below have their measurable, weighted meaning. In particular,

$$\int_{-1}^1 g(x)\,dx = \int_0^\pi g(\cos\theta)\sin\theta\,d\theta, \qquad \int_{-1}^1 g(x)\frac{dx}{y_x} = \int_0^\pi g(\cos\theta)\,d\theta$$

hold as extended identities for nonnegative measurable functions, and for complex functions whenever the displayed absolute integral is finite. The second identity requires $L^1(dx/y_x)$, rather than bare $L^1(dx)$. These statements follow first for continuous functions from monotone one-dimensional substitution, then for Borel functions by equality of the pushforward measures, and finally for completed representatives by ignoring null preimages where the Jacobian is nonzero and assigning the weighted composition zero where it vanishes. Thus no boundedness of a Hardy trace is assumed in these changes of variable.

5.3.5 Comparison with the reference energy

Proof of Theorem 5.4. Set

$$A_0 = G_0 = P^{3/4}, \qquad \rho_0(x) = (2y_x)^{3/2}, \qquad u = \log(\rho/\rho_0), \qquad \widetilde{h} = e^u.$$

Since $\rho = 2y_x M$, the function in (5.21) is $h = \sqrt{2}\,\widetilde{h}$. All its ratios are unchanged by this constant. Also

$$Z_0 = \int_0^\pi \cos\theta\ \operatorname{Im}\log A_0(e^{i\theta})\,d\theta = -\frac{3}{2}.$$

At fixed $\varepsilon > 0$, the identity

$$\Phi(r_0 e^d) = \Phi(r_0) + (r_0-1)d + r_0(e^d - 1 - d), \quad r_0 = \frac{\rho_0(t)}{\rho_0(x)}, \quad d = u(t) - u(x),$$

can be integrated termwise. Indeed (5.36) and the logarithmic bounds give the requisite absolute integrability for both ρ and ρ_0. The factor

$$\frac{y_x^2}{y_t} r_0 = \sqrt{y_x y_t}$$

is symmetric, so its linear d term integrates to zero. Symmetrization of the exponential remainder proves the exact identity

$$\mathcal{T}_\varepsilon(\rho) = \mathcal{T}_\varepsilon(\rho_0) + \mathrm{Lin}_\varepsilon(u) + \mathcal{L}_\varepsilon[h]. \tag{5.41}$$

We evaluate the reference entropy directly. Define the finite-cutoff signed quantity

$$\mathcal{R}_\varepsilon(\rho) = \frac{1}{2\pi} \iint \frac{y_x^2}{y_t} \left[\frac{\rho(t)}{\rho(x)} - 1 \right] K_\varepsilon(x,t)\, dx\, dt.$$

Then $\mathcal{T}_\varepsilon(\rho) = \mathcal{R}_\varepsilon(\rho) + \mathrm{Lin}_\varepsilon(\log \rho)$. For ρ_0, symmetrization gives

$$\begin{aligned} -\mathcal{R}_\varepsilon(\rho_0) &= \frac{1}{4\pi} \iint \left[\frac{y_x^2}{y_t} + \frac{y_t^2}{y_x} - 2\sqrt{y_x y_t} \right] K_\varepsilon\, dx\, dt \\ &= \mathcal{E}_{\mathrm{int},\varepsilon}[f_0] + V_\varepsilon, \end{aligned} \tag{5.42}$$

where

$$\mathcal{E}_{\mathrm{int},\varepsilon}[f_0] = \frac{1}{4\pi} \iint_{(-1,1)^2} [f_0(x) - f_0(t)]^2 K_\varepsilon(x,t)\, dx\, dt,$$

and

$$V_\varepsilon = \frac{1}{4\pi} \iint_{(-1,1)^2} \left[\frac{y_x^2}{y_t} + \frac{y_t^2}{y_x} - y_x - y_t \right] K_\varepsilon(x,t)\, dx\, dt.$$

All three integrands in (5.42) are nonnegative. Proposition 5.3 gives convergence of the first term to its uncut interior energy. The numerator in V_ε is

$$\frac{(y_x - y_t)^2 (y_x + y_t)}{y_x y_t},$$

and its quotient by $(x-t)^2$ equals $(x+t)^2/[y_x y_t (y_x + y_t)]$ almost everywhere. To integrate it, put $x = \cos\theta$, $t = \cos\eta$, and then $p = (\theta+\eta)/2$, $q = (\theta-\eta)/2$. The latter substitution has region

$$0 < p < \pi, \qquad |q| < \min(p, \pi - p), \qquad d\theta\, d\eta = 2\, dp\, dq.$$

It follows from two one-dimensional substitutions and Tonelli: first substitute $p = (\theta+\eta)/2$ at fixed θ, interchange p, θ, and then set $q = \theta - p$. The integrand is nonnegative throughout. As $x + t = 2\cos p \cos q$ and $y_x + y_t = 2 \sin p \cos q$, this gives

$$\begin{aligned} \iint \frac{(x+t)^2}{y_x y_t (y_x + y_t)}\, dx\, dt &= 4 \int_0^\pi \frac{\cos^2 p}{\sin p} \int_{-\min(p,\pi-p)}^{\min(p,\pi-p)} \cos q\, dq\, dp \\ &= 8 \int_0^\pi \cos^2 p\, dp = 4\pi. \end{aligned}$$

Hence $V_\varepsilon \uparrow 1$. This is also the exterior energy in (5.16), and consequently

$$-\mathcal{R}_\varepsilon(\rho_0) \longrightarrow c_*.$$

Lemma 5.7, applied to A_0, gives $\mathrm{Lin}_\varepsilon(\log \rho_0) \to -2Z_0 = 3$. We have therefore proved

$$\mathcal{T}(\rho_0) = 3 - c_* < \infty, \qquad -\mathcal{T}(\rho_0) - 2Z_0 = c_*. \tag{5.43}$$

Now $u = \log |A/A_0|^2$ lies in $L^2(d\nu)$, and $\log(A/A_0) \in H^2$. The logarithmic functional tends to $-2(Z - Z_0)$ by (5.38). The nonnegative losses in (5.41) increase to $\mathcal{L}[h]$, while $\mathcal{T}(\rho) < \infty$ by (5.33). Thus

$$\mathcal{T}(\rho) = \mathcal{T}(\rho_0) - 2(Z - Z_0) + \mathcal{L}[h], \qquad \mathcal{L}[h] < \infty.$$

Substitution in (5.35) and (5.43) proves (5.23).

The loss vanishes exactly when $h(t) = h(x)$ for almost every pair, since its kernel is strictly positive off the diagonal. Fubini then shows that h is almost everywhere constant. The outer Poisson representations in Lemma 5.5 make $\log A - \log A_0$ a constant if their boundary real parts differ by a constant. Their normalization at zero makes that constant zero. Thus equality forces $A = A_0$, or $F = P^{1/4}$.

Conversely, for $A_0 = P^{3/4}$, each fixed interior pair $\zeta, \bar{\zeta}$ is a pair of points of analytic continuation of A_0. Its interpolating quotient is in H^2: the zeros at these points cancel, and the only remaining endpoint fractional powers are bounded. Lemma 5.6 applies pointwise without needing integrated finiteness first. The finite value in (5.43), together with the integrable phase and (5.32), therefore proves $\mathcal{G}(P^{1/4}) < \infty$. Its h is constant, so (5.23) gives equality. □

For this last factor the separate terms are

$$\mathcal{A}(P^{1/4}) = \frac{1}{2}, \qquad \frac{1}{2}\int_0^\pi \operatorname{Re}\frac{P(e^{i\theta})^{1/4}}{P(e^{i\theta})^{1/4}}\,d\theta = \sqrt{2}, \qquad \mathcal{G}(P^{1/4}) = \frac{1}{2} + \sqrt{2} - c_*.$$

As further normalization checks, the Gaussian factor $F = P^{-1/2}$ gives $\rho = 1$, $\mathcal{T} = Z = 0$, and $\mathcal{B} - \mathcal{G} = 0$. For $F = 1$, $Z = -1$, $\mathcal{G} = \pi/2 - 1$, and $\mathcal{B} = \pi/2$, so $\mathcal{T} = 1$ and $\mathcal{B} - \mathcal{G} = 1$. The equality characterization above concerns the analytic factor functional; it makes no finite-degree optimizer classification.

6 An actual Jacobi node construction

We now evaluate the original unweighted cardinal energy on actual Jacobi roots. The Jacobi measure is an auxiliary device for an exact polynomial reduction; every integral defining I_n still uses ordinary dx and every cardinal has weight one. We give the calculation for fixed $-1 < \alpha \le -1/2$, and then specialize to $\alpha = -3/4$.

Proposition 6.1. *Let $X_n(\alpha)$ be the n roots of $P_n^{(\alpha,\alpha)}$, for fixed $-1 < \alpha \le -1/2$. These are distinct nodes in $(-1, 1)$, and*

$$I_n(X_n(\alpha)) = 2 - \frac{c(\alpha)}{n} + o_\alpha(n^{-1}), \tag{6.1}$$

where

$$c(\alpha) = -2\alpha(\alpha+1)\frac{\Gamma_{\mathrm{E}}(\alpha+1)\Gamma_{\mathrm{E}}(1/2-\alpha)}{\Gamma_{\mathrm{E}}(\alpha+3/2)\Gamma_{\mathrm{E}}(1-\alpha)}. \tag{6.2}$$

In particular, for $Y_n = X_n(-3/4)$,

$$I_n(Y_n) = 2 - \frac{c_*}{n} + o(n^{-1}). \tag{6.3}$$

No remainder uniform in α is asserted.

6.1 The auxiliary Gaussian identity

Throughout this section only, $q_n = P_n^{(\alpha,\alpha)}$ has the standard Jacobi normalization. Multiplying a nodal polynomial by a nonzero constant does not change its cardinals or I_n. Set

$$w_\alpha(x) = (1-x^2)^\alpha, \qquad \mu_\alpha = \int_{-1}^1 w_\alpha(x)\,dx = \frac{\sqrt{\pi}\,\Gamma_{\mathrm{E}}(\alpha+1)}{\Gamma_{\mathrm{E}}(\alpha+3/2)}.$$

The Jacobi orthogonality, squared norm, and leading coefficient are [6, Table 18.3.1]

$$\begin{aligned} h_n &= \int_{-1}^{1} w_\alpha q_n^2 \, dx = \frac{2^{2\alpha+1}\Gamma_{\mathrm{E}}(n+\alpha+1)^2}{(2n+2\alpha+1)\Gamma_{\mathrm{E}}(n+1)\Gamma_{\mathrm{E}}(n+2\alpha+1)}, \\ k_n &= [x^n]q_n = \frac{\Gamma_{\mathrm{E}}(2n+2\alpha+1)}{2^n\Gamma_{\mathrm{E}}(n+1)\Gamma_{\mathrm{E}}(n+2\alpha+1)}. \end{aligned} \tag{6.4}$$

Here the displayed norm formula is used for $n \geq 1$; $h_0 = \mu_\alpha$ is specified separately. Thus no exceptional gamma value at degree zero is used.

Orthogonality for the positive integrable weight gives n simple roots in $(-1, 1)$. Indeed, if the product of the distinct sign-changing interior roots had degree below n, its product with q_n would have one strict sign away from finitely many points, contradicting orthogonality. There must therefore be n such roots, and all roots are simple and interior.

Let ℓ_i be their cardinals, and define auxiliary weights $\omega_i = \int_{-1}^{1} w_\alpha \ell_i \, dx$. Polynomial division by q_n, followed by orthogonality, proves

$$\int_{-1}^{1} w_\alpha(x) p(x) \, dx = \sum_i \omega_i p(x_i), \qquad \deg p \leq 2n-1.$$

Applying this to ℓ_i^2 gives $\omega_i = \int w_\alpha \ell_i^2 > 0$, and $\sum_i \omega_i = \mu_\alpha$. These weights are unrelated to the ordinary Legendre Christoffel numbers used earlier.

The reproducing kernel for degree at most $n-1$ is

$$\mathcal{K}_{n-1}^{(\alpha)}(x,t) = \sum_{j=0}^{n-1} \frac{q_j(x) q_j(t)}{h_j}.$$

Subtracting the orthogonal-polynomial recurrence at x and t, then telescoping, gives

$$\mathcal{K}_{n-1}^{(\alpha)}(x,t) = \frac{k_{n-1}}{k_n h_{n-1}} \frac{q_n(x) q_{n-1}(t) - q_{n-1}(x) q_n(t)}{x - t}.$$

Gaussian exactness implies $\mathcal{K}_{n-1}^{(\alpha)}(x, x_i) = \ell_i(x)/\omega_i$. The same-parameter lowering relation [6, Eq. 18.9.17] gives at a root

$$(1 - x_i^2) q_n'(x_i) = (n+\alpha) q_{n-1}(x_i).$$

The diagonal limit of the kernel identity and (6.4) consequently yield

$$\omega_i = \frac{\kappa_n}{(1-x_i^2) q_n'(x_i)^2}, \qquad \kappa_n = (2n+2\alpha+1) h_n = \frac{2^{2\alpha+1}\Gamma_{\mathrm{E}}(n+\alpha+1)^2}{\Gamma_{\mathrm{E}}(n+1)\Gamma_{\mathrm{E}}(n+2\alpha+1)}. \tag{6.5}$$

Fixed-shift gamma asymptotics [6, Eq. 5.11.12] give

$$\kappa_n \longrightarrow 2^{2\alpha+1}. \tag{6.6}$$

All large-degree gamma arguments used here are positive, with α fixed.

6.2 Removing the nodes by Hermite interpolation

The general cardinal-square identity behind this reduction appears in [2, Theorem 2.1, Eq. (7), pp. 39–40]. We include the Hermite derivation to make the normalization explicit.

Put $a = \alpha + 1$. The Jacobi differential equation [6, Table 18.8.1] is

$$(1-x^2) q_n'' - 2(\alpha+1) x q_n' + n(n+2\alpha+1) q_n = 0. \tag{6.7}$$

At a root it implies

$$\ell_i'(x_i) = \frac{q_n''(x_i)}{2q_n'(x_i)} = \frac{ax_i}{1-x_i^2}.$$

Hermite interpolation of the constant one gives the exact identity

$$1 = \sum_i [1 - 2\ell_i'(x_i)(x - x_i)]\ell_i(x)^2.$$

The right side has degree at most $2n-1$, equals one at every node, and has zero derivative there; its difference from one has n double zeros and is therefore zero.

Define the polynomial of degree at most $n-1$

$$p_n(x) = \int_{-1}^{1} \frac{q_n(x) - q_n(t)}{x - t}\, w_\alpha(t)\, dt. \tag{6.8}$$

Gaussian exactness in t gives $p_n(x) = q_n(x)\sum_i \omega_i/(x-x_i)$ away from the roots, and everywhere by polynomial continuation. Substituting (6.5) into the Hermite identity proves

$$\sum_i \ell_i(x)^2 - 1 = \frac{2a}{\kappa_n}[xq_n(x)p_n(x) - \mu_\alpha q_n(x)^2]. \tag{6.9}$$

For example, the sum giving its right side reduces by $x_i/(x-x_i) = x/(x-x_i) - 1$ and $\sum_i \omega_i = \mu_\alpha$. Integration in the original ordinary measure now gives

$$2 - I_n(X_n(\alpha)) = \frac{2a}{\kappa_n} D_n, \qquad D_n = \mu_\alpha \int_{-1}^{1} q_n^2\, dx - \int_{-1}^{1} xq_n p_n\, dx. \tag{6.10}$$

6.3 A global bound and two weak limits

To control the singular endpoints independently of the interior asymptotic, put $\Lambda_n = n(n + 2\alpha + 1) > 0$ and

$$E_n(x) = q_n(x)^2 + \frac{1-x^2}{\Lambda_n} q_n'(x)^2.$$

Direct differentiation using (6.7) gives

$$E_n'(x) = \frac{2(2\alpha+1)x}{\Lambda_n} q_n'(x)^2. \tag{6.11}$$

Since $-1 < \alpha \le -1/2$, this is nonpositive for $0 \le x \le 1$; parity handles the other half interval. Thus $\sup q_n^2 \le E_n(0)$.

The finite binomial representation [6, Eq. 18.5.8] gives

$$q_{2m}(0) = (-1)^m \frac{\Gamma_{\mathrm{E}}(2m + \alpha + 1)}{2^{2m}\Gamma_{\mathrm{E}}(m+1)\Gamma_{\mathrm{E}}(m + \alpha + 1)}, \qquad q_{2m+1}(0) = 0. \tag{6.12}$$

The finite sum collapses by taking the degree-$2m$ coefficient of $(1-z)^{2m+\alpha}(1+z)^{2m+\alpha} = (1 - z^2)^{2m+\alpha}$, with the factor 2^{-2m}. Stirling's formula on the positive ray gives $q_{2m}(0) = O_\alpha(m^{-1/2})$. For odd n, the derivative formula [6, Eq. 18.9.15] gives

$$q_n'(0) = \frac{n + 2\alpha + 1}{2} P_{n-1}^{(\alpha+1,\alpha+1)}(0) = O_\alpha(n^{1/2}),$$

by the same center formula with the shifted fixed parameter. The even derivatives at zero vanish. Since $\Lambda_n \asymp_\alpha n^2$, we obtain

$$\sup_{n\ge 2} \sup_{-1\le x\le 1} |\sqrt{n}\, q_n(x)| < \infty. \tag{6.13}$$

Let $g_n = \sqrt{n}\, q_n$. On every compact angular subinterval, the Darboux formula [6, Eq. 18.15.4_5] gives

$$g_n(\cos\theta) = \frac{2^{\alpha+1/2}}{\sqrt{\pi}} (\sin\theta)^{-\alpha-1/2} \cos\left[(n+\alpha+\tfrac{1}{2})\theta - (\alpha+\tfrac{1}{2})\frac{\pi}{2}\right] + O_{\alpha,\delta}(n^{-1}). \tag{6.14}$$

For an L^1 angular amplitude, its oscillatory integral tends to zero: approximate it in L^1 by interval step functions, use the bound 2/|frequency| for the integral of an exponential on each interval, and bound the approximation error by its L^1 norm. Applying this to (6.14) and its square gives

$$g_n \rightharpoonup^* 0, \qquad g_n^2 \rightharpoonup^* v_\alpha(x) := \frac{2^{2\alpha}}{\pi}(1-x^2)^{-\alpha-1/2} \quad \text{in } \sigma(L^\infty, L^1). \tag{6.15}$$

Indeed, given an arbitrary $L^1(dx)$ test, first restrict it to $[-1+\delta, 1-\delta]$ and use the angular substitution there. The errors in (6.14) are uniform on this interval. The omitted contributions are bounded by a common constant times the test's L^1 mass in the endpoint regions, using (6.13). The limit density v_α is also bounded because $\alpha \le -1/2$. Let $\delta \downarrow 0$. This proves the limits against all L^1 tests, including integrable singular functions.

6.4 The finite Hilbert transform and its truncations

Define the raw finite Hilbert transform

$$W_\alpha(x) = \mathrm{PV} \int_{-1}^{1} \frac{w_\alpha(t)}{x-t}\, dt,$$

with symmetric truncations $|x-t| > \varepsilon$. They satisfy the uniform integrable estimate

$$|W_{\alpha,\varepsilon}(x)| \le C_\alpha[1 + (1-x)^\alpha + (1+x)^\alpha]. \tag{6.16}$$

Near $x = 1$, put $u = 1-x$, $v = 1-t$, and write the weight as $v^\alpha H(v)$, where H is positive and C^1 on a fixed endpoint neighborhood. On $[u/2, 3u/2]$, the constant $u^\alpha H(u)$ cancels across the symmetric truncation; the remaining quotient is bounded by $Cu^{\alpha-1}$ over a set of length at most u. On $0 < v < u/2$, integrate Cv^α/u, and on $v > 3u/2$ integrate $Cv^{\alpha-1}$. Each contribution is $O(u^\alpha)$ for $-1 < \alpha < 0$, including when the truncation removes the whole middle interval. The separated part contributes $O(1)$. The other endpoint is the same; on compact interiors, subtraction of the local weight value gives a uniform bound. This proves (6.16).

On compact interiors, subtracting the Taylor expansion of the weight justifies the locally uniform finite-part derivative

$$W'_\alpha(x) = \lim_{\varepsilon\downarrow 0} \left[-\int_{|t-x|>\varepsilon} \frac{w_\alpha(t)}{(x-t)^2}\, dt + \frac{w_\alpha(x-\varepsilon) + w_\alpha(x+\varepsilon)}{\varepsilon} \right].$$

Integrate the derivative in t of $(1-t^2)w_\alpha(t)/(x-t)$ over the truncated interval. The endpoint terms vanish since $\alpha + 1 > 0$; the two cut terms are $2(1-x^2)w_\alpha(x)/\varepsilon + O(\varepsilon)$. Using $1 - t^2 = (1-x^2) + 2x(x-t) - (x-t)^2$, the integral of this derivative also equals

$$(1-x^2)\int_{|x-t|>\varepsilon} \frac{w_\alpha(t)}{(x-t)^2}\, dt - 2\alpha x W_{\alpha,\varepsilon}(x) + (2\alpha+1)\int_{|x-t|>\varepsilon} w_\alpha(t)\, dt.$$

Cancellation with the finite-part derivative therefore gives

$$(1-x^2)W'_\alpha(x) + 2\alpha x W_\alpha(x) = (2\alpha+1)\mu_\alpha. \tag{6.17}$$

Evenness of the weight gives $W_\alpha(0) = 0$. Solving the first-order equation yields

$$W_\alpha(x) = (2\alpha+1)\mu_\alpha(1-x^2)^\alpha \int_0^x (1-s^2)^{-\alpha-1}\, ds. \tag{6.18}$$

The inner integral is bounded on $[-1, 1]$ for the present negative α, so $W_\alpha \in L^1$, consistently with (6.16). At $\alpha = -1/2$ the formula gives $W_\alpha = 0$.

6.5 The second-kind oscillation as an integrable kernel

The divided-difference definition (6.8) gives

$$p_n(x) = q_n(x)W_\alpha(x) - \mathrm{PV}\int_{-1}^{1} \frac{q_n(t)w_\alpha(t)}{x-t}\,dt.$$

Hence

$$D_n = \int_{-1}^{1} q_n(x)^2[\mu_\alpha - xW_\alpha(x)]\,dx + J_n, \tag{6.19}$$

where

$$J_n = \int_{-1}^{1} xq_n(x)\,\mathrm{PV}\int_{-1}^{1} \frac{q_n(t)w_\alpha(t)}{x-t}\,dt\,dx.$$

The precise symmetrized formula is

$$J_n = \iint_{(-1,1)^2} L_\alpha(x,t)q_n(x)q_n(t)\,dx\,dt, \qquad L_\alpha(x,t) = \frac{xw_\alpha(t) - tw_\alpha(x)}{2(x-t)}. \tag{6.20}$$

We check both the kernel integrability and the passage from truncated integrals.

First, $L_\alpha \in L^1((-1,1)^2)$, since

$$2L_\alpha = w_\alpha(t) + t\frac{w_\alpha(t) - w_\alpha(x)}{x-t}.$$

The first term is integrable. For the divided difference, compact-interior regions are bounded. At a common endpoint, write the weight as $u^\alpha H(u)$ with bounded C^1 factor H. The difference of the H factors is integrable. On $0 < u < v < \eta$, the power difference satisfies

$$\int_0^\eta \int_0^v \frac{u^\alpha - v^\alpha}{v-u}\,du\,dv = \left(\int_0^\eta v^\alpha\,dv\right)\left(\int_0^1 \frac{r^\alpha - 1}{1-r}\,dr\right) < \infty, \tag{6.21}$$

by $u = rv$. Both factors are finite for $-1 < \alpha < 0$. Endpoint regions separated from each other or from a compact interval have denominator bounded below and use $w_\alpha \in L^1$. A finite partition of the square proves the claim.

Second, for fixed n the truncated identity reads

$$H_\varepsilon(q_n w_\alpha) = q_n W_{\alpha,\varepsilon} - p_{n,\varepsilon},$$

where H_ε denotes the raw truncated integral and $p_{n,\varepsilon}$ is the same truncation of (6.8). The divided difference of the fixed polynomial is bounded on the square, and the weight mass of a removed interval of length 2ε tends to zero uniformly. Thus $p_{n,\varepsilon} \to p_n$ uniformly. The bound (6.16) gives dominated convergence for the outer integral containing $q_n^2 W_{\alpha,\varepsilon}$. Fubini is legitimate at each positive cutoff. After symmetrizing, the integrand is dominated by the integrable $|L_\alpha|$ times a fixed polynomial bound. Dominated convergence proves (6.20).

Multiplying that formula by n produces the factors $g_n(x)g_n(t)$. Approximate the fixed L^1 kernel by a finite sum of rectangle indicators. On every rectangle, the integral separates and tends to zero by (6.15). The error is bounded by $\sup_n \|g_n\|_\infty^2$ times the L^1 approximation error. We conclude that

$$nJ_n \longrightarrow 0. \tag{6.22}$$

Thus no pointwise asymptotic for p_n, and no boundedness theorem for a weighted Hilbert transform, is needed.

6.6 Evaluation of the coefficient

Proof of Proposition 6.1. The multiplier $\mu_\alpha - xW_\alpha(x)$ is in L^1. Pair it with the second limit in (6.15) and use (6.19)–(6.22):

$$\lim_n nD_n = \frac{2^{2\alpha}}{\pi}\int_{-1}^{1}(1-x^2)^{-\alpha-1/2}[\mu_\alpha - xW_\alpha(x)]\,dx. \tag{6.23}$$

Put

$$V_\alpha = \int_{-1}^{1}(1-x^2)^{-\alpha-1/2}\,dx = \frac{\sqrt{\pi}\,\Gamma_{\mathrm{E}}(1/2-\alpha)}{\Gamma_{\mathrm{E}}(1-\alpha)}.$$

Using (6.18) and parity gives

$$\begin{aligned}
&\int_{-1}^{1} x(1-x^2)^{-\alpha-1/2}W_\alpha(x)\,dx \\
&\quad= 2(2\alpha+1)\mu_\alpha\int_0^1 \frac{x}{\sqrt{1-x^2}}\int_0^x (1-s^2)^{-\alpha-1}\,ds\,dx \\
&\quad= 2(2\alpha+1)\mu_\alpha\int_0^1 (1-s^2)^{-\alpha-1}\sqrt{1-s^2}\,ds \\
&\quad= (2\alpha+1)\mu_\alpha V_\alpha.
\end{aligned}$$

The interchange follows first by Tonelli for the nonnegative integrand after factoring out the constant. Its final value is finite, proving absolute integrability also when the constant is negative. Therefore the integral in (6.23) is $-2\alpha\mu_\alpha V_\alpha$.

Finally (6.6) and (6.10) yield

$$\lim_n n(2 - I_n(X_n(\alpha))) = \frac{2(\alpha+1)}{2^{2\alpha+1}}\,\frac{2^{2\alpha}}{\pi}(-2\alpha\mu_\alpha V_\alpha) = -\frac{2\alpha(\alpha+1)}{\pi}\mu_\alpha V_\alpha.$$

The beta integrals for μ_α and V_α, using $t = x^2$, have positive parameters; the beta–gamma identity [6, Eq. 5.12.1] gives (6.2). For $\alpha = -3/4$, the positive-real recurrence $\Gamma_{\mathrm{E}}(s+1) = s\Gamma_{\mathrm{E}}(s)$ gives

$$c(-3/4) = \frac{3}{8}\,\frac{\Gamma_{\mathrm{E}}(1/4)\Gamma_{\mathrm{E}}(5/4)}{\Gamma_{\mathrm{E}}(3/4)\Gamma_{\mathrm{E}}(7/4)} = \frac{\Gamma_{\mathrm{E}}(1/4)^2}{8\Gamma_{\mathrm{E}}(3/4)^2} = c_*.$$

This proves (6.3). Since $c_* > 0$, these actual legal nodes have $I_n(Y_n) < 2$ for all sufficiently large n. In particular $m_n \le I_n(Y_n)$ gives $\liminf_n n(2-m_n) \ge c_*$, with the original ordinary dx, unit-cardinal normalization throughout. □

7 Proof of the sharp asymptotic

Write $\mathcal{X}_n = \{X \subset [-1,1] : |X| = n\}$ for the class of admissible node sets. We first establish the following assertion for arbitrary sublevel configurations:

$$n_j \to \infty, \quad X_j \in \mathcal{X}_{n_j}, \quad I_{n_j}(X_j) \le 2 \quad\Longrightarrow\quad \limsup_j n_j(2 - I_{n_j}(X_j)) \le c_*. \tag{7.1}$$

The finite all-node inequality bounds the nonnegative scalars on the right by $D/2$. Select a subsequence attaining their limsup. Refine it by the analytic coefficient extraction, the weak phase extraction and the countable collection of bulk kernel extractions above. These refinements preserve the chosen scalar limit. On this common subsequence the reciprocal analytic limit is uniquely $G = P/F$, and the full scalar limit is (4.20). No phase from a separately chosen subsequence is inserted.

The top-mode estimate gives

$$\liminf_j \Delta_{n_j} \geq \mathcal{G}(F), \qquad 0 \leq \mathcal{G}(F) \leq D.$$

The entropy comparison applies to precisely this actual limit class, $A, G \in H^2$, $AG = P^{3/2}$ and finite $\mathcal{G}(F)$, and yields

$$\mathcal{B}(F) - \mathcal{G}(F) \leq c_*.$$

Equation (4.23) therefore proves

$$\limsup_j n_j(2 - I_{n_j}(X_j)) \leq \mathcal{B}(F) - \liminf_j \Delta_{n_j} \leq \mathcal{B}(F) - \mathcal{G}(F) \leq c_*,$$

which establishes (7.1). Each compact-interior asymptotic has been used before endpoint exhaustion, with its corresponding one-sided bound or full integrable majorant.

We now pass from configurations to the unrestricted infimum. The class $\mathcal{X}_n$ is nonempty, and $\sum_i \ell_i = 1$ gives $I_n \geq 2/n$. Thus m_n is finite. For each n choose $X_n \in \mathcal{X}_n$ satisfying

$$m_n \leq I_n(X_n) \leq m_n + n^{-2}. \tag{7.2}$$

This is the defining approximation property of an infimum and does not assume attainment.

The Jacobi construction gives legal node sets Y_n satisfying

$$I_n(Y_n) = 2 - \frac{c_*}{n} + o(n^{-1}).$$

Since $c_* > 0$, eventually $m_n \leq I_n(Y_n) \leq 2 - c_*/(2n)$. Equation (7.2) then places X_n in the strict sublevel $I_n(X_n) < 2$ for all sufficiently large n. Furthermore,

$$0 \leq n(2 - m_n) - n(2 - I_n(X_n)) = n[I_n(X_n) - m_n] \leq n^{-1}.$$

Apply (7.1) to every limsup-selecting subsequence of these near-infima. It follows that

$$\limsup_{n\to\infty} n(2 - m_n) \leq c_*.$$

Conversely $m_n \leq I_n(Y_n)$ gives

$$\liminf_{n\to\infty} n(2 - m_n) \geq c_*.$$

Consequently $n(2 - m_n) \to c_*$, and rearranging $m_n = 2 - n(2 - m_n)/n$ proves the expansion in Theorem 1.1. At $n = 1$ the sole cardinal is one, so $I_1 = m_1 = 2$; this finite prefix affects no limit. □

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